

Human Capital, Competition and Mobility in the Managerial Labor Market*

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Abstract

We estimate a search model of managerial careers to quantify the relative importance of human capital accumulation (both general and firm-specific), managerial bargaining power, and imperfect labor market competition in shaping compensation and mobility in the market for US corporate executives. The composition of human capital is career-dependent and varies widely across managers: over tenure, firm-specific capital is the greatest driver of wage growth, whereas over experience in the labor market, job search and competition dominate. Firm-specific capital can help explain the high rate of internal CEO hires and low observed cross-firm CEO mobility. We further show that labor market competition (relative to pure bargaining power) makes up the majority of realized CEO surplus capture, and that firm-specific human capital positively interacts with competition in determining CEOs' shares of rents because it raises the match-specific quality between the firm and manager.

Keywords: CEOs, the market for CEOs, executive pay, executive mobility, on-the-job search, structural estimation, human capital accumulation, firm-specific human capital

*This paper is a work in progress and results are subject to change, all errors are our own.

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1. Introduction

In this paper, we quantify the relative importance of human capital accumulation, both general and firm-specific, managerial bargaining power, and imperfect labor market competition in determining compensation and mobility in the market for US corporate executives. We do so by posing and estimating a structural model of managerial careers, and our paper contributes to the literature in three ways.

First, decomposing managerial human capital into general and firm-specific components has important implications for the study of executives: prevailing theories rationalize the recent rise in CEO pay via a premium on general, portable CEO skill (e.g., [Gabaix and Landier, 2008](#); [Murphy and Zabojnik, 2007](#)).¹ A shared feature of these models is that competition for generalist CEOs induces higher wages via an outside option channel, and while these models may not be intended to explain managerial mobility, a direct empirical implication is that managers should move frequently across firms. Yet, as shown in [Cziraki and Jenter \(2024\)](#), around 72% of new CEOs at the largest companies in the US are internally promoted, and only a small percentage of external hires are poached CEOs ([Graham et al., 2020](#); [Cziraki and Jenter, 2024](#)), suggesting that firm-specific managerial skill may be more important than previously considered.²

Second, understanding the contributions of pure bargaining power and imperfect labor market competition to managerial surplus capture informs the large literature on CEO bargaining power and influence on compensation policy.³ Competition for CEO services can lead to surplus capture even if CEOs' pure *ex ante* bargaining power is low; explicitly modeling the labor market for managers is essential for quantifying agency frictions in the pay-setting process.

¹[Gabaix and Landier \(2008\)](#) is an example of a competitive assignment model that allow for complementarity between CEO skill and firm size, which helps explain the rise in CEO pay. [Murphy and Zabojnik \(2007\)](#), in a separate but related setting, shows that general managerial skill improves managers' outside options, which can also help rationalize the increase in CEO pay.

²[Cziraki and Jenter \(2024\)](#) show that 73% of new CEOs at S&P500 firms are internal hires from 1994-2012, and a large proportion of external hires are known to the Board (either a former executive, or a Board member). [Graham et al. \(2020\)](#) show that 68% of new CEOs are internal hires (a current or previous officer of the firm) for a fuller sample of NYSE/Amex firms from 1933-2011.

³See, e.g., [Core et al. \(1999\)](#); [Bertrand and Mullainathan \(2001\)](#); [Bebchuk and Fried \(2003\)](#); [Fahlenbrach \(2009\)](#); [Bebchuk et al. \(2011\)](#); [Morse et al. \(2011\)](#); [Taylor \(2013\)](#); [Coles et al. \(2014\)](#).

Third, we show that the managerial human capital and surplus capture channels interact in determining mobility and compensation. Firm-specific human capital accumulation increases the manager’s match quality with the firm over tenure, which can help explain the low rate of cross-firm mobility. However, it also has a positive impact on realized managerial surplus capture: more firm-specific skill raises the opportunity cost of losing the manager, making the firm more likely to match attractive outside offers. As such, firm-specific human capital can help rationalize simultaneously low mobility and high realized bargaining power of managers.

Our structural model is a comprehensive, tractable characterization of managerial careers, (both non-CEO and CEO), incorporating general and firm-specific human capital accumulation, managerial bargaining power, labor market competition, mobility (including both internal and external promotion opportunities), search frictions, and executive and firm heterogeneity. Unlike the existing literature which generally focuses on firm demand for managerial talent, we model the executive’s career through the lens of on-the-job search (Bagger et al., 2014), in which these forces jointly determine mobility and compensation growth throughout a manager’s career.⁴

Managers in the model may be employed as a (non-CEO) executive or as the CEO of the firm.⁵ Over their careers, executives may receive job offers — to move horizontally, be promoted internally to CEO, or to be promoted externally to CEO; the arrival rates of these job offers differ and are estimated in the data. CEOs may receive job offers to be CEOs at other firms.

We follow an important strand of the search literature (Postel-Vinay and Robin, 2002; Bagger et al., 2014) and model managerial compensation as a piece-rate contract: managers receive a portion $R \in [0, 1]$ of their contribution to firm output. When a manager receives an attractive outside offer, the incumbent and poaching firm may bargain over the executive’s services (in the spirit of Rubinstein, 1982). Firm-switching events occur when the poaching firm values the manager more than the incumbent. On-the-job search leads to stochastic, discrete increases in

⁴Indeed, while firm-specific human capital can help explain preference for insiders and low CEO mobility (Cziraki and Jenter, 2024), its impact is not immediately separable from search frictions inhibiting the movements of CEOs across firms (such as a preference for internal CEO promotions, He and Schroth, 2024). Moreover, differences in wages across executives and firms may be attributable to time-invariant (possibly unobserved) heterogeneity, either across managers or positions, or both.

⁵To limit confusion, throughout the paper we refer to non-CEO executives as “executives” and CEOs as “CEOs.”

pay as firms bargaining over managerial services, even if the manager ultimately stays in their current position.

We adapt the setting in [Bagger et al. \(2014\)](#) to allow for firm-specific human capital (in addition to general) and for both external and internal promotion of executives. Importantly, we let search rates and executive bargaining power differ across internal and external promotion opportunities, which enables us to distinguish a preference for insiders ([He and Schroth, 2024](#)) from the impact of firm-specific human capital on internal vs. external executive mobility.

One insight from the model is that firm-specific human capital accumulation leads to increased match-specific productivity between the firm and manager over the manager’s tenure. This makes the manager less likely to be tempted away by poaching offers as they advance at the incumbent (decreasing job-to-job transitions). However, firm-specific human capital increases the likelihood that the incumbent firm is willing to match attractive outside offers, precisely because of the increased match-specific productivity. We show theoretically that cross-firm mobility decreases with firm-specific human capital accumulation, whereas within-firm mobility (contract renegotiation due to outside offers) increases.

Our model also provides closed-form solutions for managerial contracts, and we demonstrate that, all else being equal, internally promoted CEOs receive a smaller share of rents compared to externally-hired candidates, with poached CEOs receiving the largest share. This outcome holds regardless of the level of firm-specific human capital. The reason lies in the differing outside options available to these three types of managers. Consequently, our model predicts that CEO contracts are path-dependent, shaped by whether the CEO was hired internally or externally.

We estimate the model on a rich panel of managerial careers spanning 1992-2023 (Execu-comp), combined with manually-collected data on managers’ tenures at firms and their experience in the executive labor market. This allows us to track experience, tenure, compensation, and mobility over the working life of a manager. Transitions of managers (non-CEO and CEO) across firms, and the resultant impacts on wages and mobility, allow us to separately identify general and firm-specific components of executive human capital.

Our estimation produces several sets of results. First, we decompose managerial human capital into its general and firm-specific components. This is a standing open question in corporate governance and the answer has important consequences for the literature on executive compensation. We find that, on average, CEO human capital is 54.4% general and 45.6% firm-specific. However, given that managers lose firm-specific capital in job-to-job transitions, the proportions vary with levels of labor market experience; for example, firm-specific capital makes up roughly 49.9% of the total for managers in the first 10 years of their career, whereas this proportion decreases to about 40.6% after 25 years of managerial experience (the difference can be explained by job-to-job transitions early on in CEO careers). Similarly, firm-specific capital has a relatively larger impact on compensation across tenure (accounting for about 45% of cumulative wage growth within tenure) than across labor market experience (about 25%).

We also analyze how firm-specific human capital impacts the hiring of internal vs. external CEOs. By simulating a counterfactual with no firm-specific skill and comparing to the baseline estimation, we find that removing firm-specific human capital lowers the proportion of internal CEO hires by about 15 percentage points (a decrease from 63% to 48%, or nearly a 24% fall), with about two thirds of the decrease going to poached CEOs (the remainder to poached executives). While search frictions and strategic cross-firm competition explain the majority of the observed rate of internal vs. external hiring, this result contributes to the understanding of mobility in the CEO labor market, stressing the importance of incorporating both firm-specific human capital and imperfect labor market competition ([Cziraki and Jenter, 2024](#)).

Our second set of results concerns realized managerial surplus capture and labor market competition. Our estimation allows us to decompose CEO rent sharing into pure CEO bargaining power and labor market competition pushing CEO wages up (which is inspired by [Cahuc et al., 2006](#)). We find that, on average, CEOs capture about 56% of rents, which is closely in line with estimates from papers in the literature. However, our estimate of pure CEO bargaining power is only 13.6%, suggesting that about 75% of realized CEO surplus capture is driven by labor market competition.

We also find that realized CEO surplus capture varies depending on if the CEO was internally promoted or externally poached, with poached CEOs commanding the greatest share of rents: internal CEOs capture 51.5% of surplus, poached CEOs 68.6%.⁶ These results contribute to the literature on agency frictions and bargaining power in executive compensation, showing that endogenizing labor market competition for executive services is crucial for understanding patterns in executive compensation.

Lastly, we show that firm-specific human capital impacts realized CEO surplus capture. We study how CEOs' shares of rents evolve over the tenure of the CEO in the baseline estimation and a simulated counterfactual without firm-specific capital, and show that firm-specific capital nearly doubles the CEO's share of rents. This stems from the theoretical result concerning how firm-specific human capital raises the firm's willingness to match attractive outside offers. Taken together, firm-specific human capital can help explain how market competition for CEO services can lead to low cross-firm mobility and high CEO surplus capture.

The paper is organized as follows. The rest of this section discusses our paper's place in the literature. Section 2 summarizes our estimation sample. Section 3 introduces the theoretical model. Section 4 discusses our estimation and identification strategies and Section 5 gives the estimated parameters. Section 6 analyzes CEO bargaining power and labor market competition. Section 7 analyzes the role of firm-specific human capital (and other channels) in determining mobility and compensation. Finally, Section 8 concludes. Model proofs and additional details are in Appendix A, and estimation details are contained in B.

1.1. Literature Review

Our paper lies in the intersection of the on-the-job search literature in labor economics, and the study of executives in corporate finance. On the labor side, our paper is closely related to preceding work concerning on-the-job search, bargaining, and human capital. Starting with [Postel-Vinay and Robin \(2002\)](#) and [Cahuc et al. \(2006\)](#), the authors develop and microfound a workhorse

⁶This result stems from our theoretical result that poached CEOs receive strictly larger contracts than internally-hired CEOs.

structural model of careers with on-the-job search. [Bagger et al. \(2014\)](#) continues by adding general human capital accumulation to the model (among other features), and study its effect on worker careers.

We expand these models in several key ways. First, we allow the worker to accumulate both general and firm-specific capital, and show that firm-specific skill has previously unstudied impacts on labor market mobility and renegotiation over worker careers.⁷ Second, we allow the worker to possibly experience both internal and external mobility throughout their career, by allowing managers to be promoted to CEO at their current firm, as well as receive job offers to be CEO at other firms.

Our paper also relates to [Jarosch \(2023\)](#), who studies how unemployment and unemployment risk interact with (the loss of) human capital, leading to endogenously-different stocks of human capital at each end of the job ladder. In our paper, movement along the job ladder also impacts a manager's human capital, but through the manager choosing whether to forgo firm-specific capital if outside job offers are attractive enough to afford the loss in skill.

On the corporate finance side, the relative importance of general and firm-specific skill, or more specifically how this relative importance can explain CEO wage growth over time, has been extensively studied. Indeed, a large literature has arisen which stresses the importance of transferable executive skill in explaining the large observed increase in CEO wages over time.⁹ However, to our knowledge, no paper has explicitly attempted to directly quantify the weights of general and firm-specific human capital in managerial skill.

[Murphy and Zabojnik \(2007\)](#) (and the related [Murphy and Zabojnik, 2004](#)) model the firm's choice of an internal vs. an external candidate, and show that, in a market with an elastic supply of executive labor, a larger importance on general executive skill (relative to firm-specific)

⁷As pointed out by [Lazear \(2009\)](#) and [Bagger et al. \(2014\)](#), firm-specific human capital has received relatively less attention than general in the labor economics literature, primarily because of the focus on rank-and file workers.⁸ However, firm-specific human capital is likely to be more important in the executive labor market, where, for example, fostering corporate culture ([Graham et al., 2022](#)), or efficiently deploying the firm's factors of production are important skills that may not transfer.

⁹This is a large, important recent literature in corporate finance, for example, [Gabaix and Landier \(2008\)](#); [Terviö \(2008\)](#); [Edmans et al. \(2009\)](#); [Murphy and Zabojnik \(2007\)](#); [Custódio et al. \(2013\)](#); [Frydman \(2019\)](#)

can rationalize the observed increases in executive pay that have been observed in the data. As [Custódio et al. \(2013\)](#) point out, a direct implication of a competitive executive labor market is that firm-specific human capital receives a lower premium in wages, because firm-specific human capital does not improve the executive's outside option.

However, as pointed out by [Cziraki and Jenter \(2024\)](#), the labor market for CEOs (especially at large firms) seems to be highly frictional, with little movement of executives across firms as wages change and with executives rarely using their outside option; given these frictions, they suggest that the market for CEOs may involve firm-specific capital and frictions, in which the firm and CEO bargain to split the surplus generated by the match.

In a model of the executive labor market with on-the-job search, managerial bargaining power, in which incumbents and poachers compete for managerial talent, the above predictions of [Murphy and Zabojnik \(2007\)](#) and [Custódio et al. \(2013\)](#) are not necessarily true. While the manager's utilization of the outside option may decrease with tenure, the importance of the *renegotiation option*, in which managers use outside offers to improve their incumbent contract, become relatively more important. Moreover, our estimates contribute to understanding of the role of labor market competition in determining CEO rent shares, which contributes to the structural literature on the executive labor market ([Taylor, 2010, 2013](#); [Page, 2018](#); [Barry, 2023](#); [Lyman, 2024](#)).

Several papers have measured the importance of general and firm-specific human capital in high-skilled industries ([Gao et al., 2021](#); [Ma et al., 2023](#)). These papers allow the worker to accumulate human capital via learning-by-doing, and study how this shapes mobility choice across career. Our paper complements these studies by (i) focusing on the managerial labor market at US public companies more generally, and (ii) explicitly studying how firm-specific capital shapes mobility paths across careers. Indeed, our analysis on the role of job search in determining managerial rent shares is similar in flavor to the measure of "bargaining capital" in these papers.

2. Data

We estimate our model on a rich panel dataset covering executive careers at US public firms from 1992 to 2023. We supplement data from Execucomp with manually-collected data on when managers first joined their firm, extending the data work from [Gayle et al. \(2015\)](#). Doing so allows us to accurately define managerial tenure and measure firm-specific human capital.

We classify each manager in our sample as being employed in an executive (non-CEO), a CEO position, or unattached. Unattachment refers to managers who stop working at a firm in our sample for at least one year, but begin work at another public firm later in their career. We label managers who drop out of the sample permanently as retired.¹⁰

Our measure of compensation is Execucomp’s TDC1, which includes each year’s salary, bonus, stock grants, option grants (valued using Black-Scholes), long-term incentive payouts, and other non-equity incentive payouts. This measure of pay clearly contains (long-term) incentive pay, which is outside of our model. Following arguments in [Taylor \(2013\)](#), we assume that the wage level set each year induces the manager to deploy their human capital at the firm and produce, and any negotiated future incentive today is incorporated into the current piece-rate.

We identify each labor market event for managers in our sample. Executives can be internally promoted, externally promoted, or move horizontally to be an executive at another firm. CEOs can move horizontally to other CEO positions. All managers can enter unattachment or retire. Unattached managers can enter executive or CEO employment.

Table 1 displays summary statistics (Panel A) and labor market transitions statistics (Panel B) for our estimation sample. Wages are deflated to 2010 dollars, and statistics are displayed conditional on employment state (executive, CEO, or unattached). The median CEO in our sample receives about 3 million dollars, with the median executive receiving about 959 thousand dollars. Wage growth is on average about 10.4% (13.4%) per year for CEOs (executives).

¹⁰The data does not allow us to perfectly separate unattachment and retirement. Indeed, any manager who initially enters unattachment (and is still looking for managerial positions), but ultimately never shows up again in Execucomp will be labeled as retirement. As such, it is likely that our separation rate η is biased downward, and our retirement rate μ is biased upwards. This has implications for the stock of managers that firms consider for unfilled positions. However, this does not impact within- and across-job bargaining, mobility and wages.

Less than one percent of managers switch firms each year. External CEO hires are rare relative to internal promotions (575 total external CEO hires relative to 4,568 promotions). These differences could reflect both the importance of firm-specific human capital and different search probabilities for internal and external candidates. An important aspect of our identification and estimation will be separating these two sources. There are more executive-to-executive transitions. Our identification strategy relies on changes in wages around job-switching events (conditional on labor market experience and position fixed effects), so these transitions prove useful.

3. Model

3.1. Environment

The labor market consists of firms and managers. Time is continuous. For a given manager, we let t denote their total experience in the managerial labor market and $\tau \leq t$ denote experience at their current firm (i.e. tenure). At any point in time, managers may find themselves in one of three employment states. If currently employed in a managerial position, managers may hold a CEO position (C) or a non-CEO position (E), where we refer to the latter type of manager as an “executive.” Otherwise, managers may be “unattached” (U), in which case they do not currently hold a position in the managerial labor market. Employed managers, both CEOs and executives, are compensated in proportion to their contribution to firm output; output is influenced by managers’ position-specific productivity and their level of human capital.

Production and human capital. A manager’s total human capital (in logs) is given by:

$$h(t, \tau) = a + g(t) + k(\tau) \tag{1}$$

The parameter $a \sim N(0, \sigma_a^2)$ is a manager-specific skill parameter reflecting permanent differences in individual ability. The functions $g(t)$ and $k(\tau)$ denote general and firm-specific human capital accumulated through work experience. Importantly, any time a manager leaves their firm, their general human capital $g(t)$ is retained while their firm-specific capital $k(\tau)$ is not. In other words,

if a manager takes a position with another firm or enters unattachment, $k(\tau)$ resets to $k(0) = 0$.

Each manager-position match generates output $Y(t, \tau, p)$, the log value of which is:

$$y(t, \tau, p) = p + h(t, \tau), \quad p \sim F \quad (2)$$

where $p \in [p_{min}, p_{max}]$ reflects the manager's position-specific match productivity, summarizing the manager's fitness for a given job. As p is position-specific, it is redrawn from distribution F any time a manager switches jobs, whether within or across firms. This assumption captures the fact that managerial productivity may differ both across firms and within-firm: a given manager may be more productive as a non-CEO executive than as CEO at their firm.

As we will later show, the quantities p and $k(\tau)$ play a critical role in managers' decision over whether to accept a competing job offer or remain in their current position. When considering job transitions, the potential gains in match productivity realized upon transition from a low- p to a high- p position must be weighed against the firm-specific capital $k(\tau)$ lost during the transition. As such, managers with high levels of human capital accrued at the incumbent firm may be inclined to reject offers from competing firms, even if the offer entails a high level of match productivity.

When a manager is not employed as an executive or CEO and is instead unattached, their tenure τ is fixed at zero and firm-specific human capital does not accumulate. We allow experience t to accumulate while unattached. The nature of the data on executive wages and mobility necessarily limits us to analyzing publicly-listed firms. Given the nature of the executive labor market, it is likely that unattached executives are accumulating experience, for example by managing a private firm. Upon entering (observable) managerial employment, a manager's tenure and firm-specific capital immediately begin to accumulate.

Retirement, unattachment and on-the-job search. Employed managers engage in on-the-job search and may be contacted by a potential employer about a new position at any time. Managers' current position type impacts the types of offers they may field. For executives, three

possible offers may be received: internal promotion to CEO, external promotion to CEO, or a lateral move to an executive position with another firm. Opportunities for an internal promotion arrive at rate λ_0 . Relative to cross-firm transitions, internal promotions do not entail losses in firm-specific capital: if in an executive accepts an internal promotion to CEO, $k(\tau)$ is retained and continues to accumulate.

External promotions arrive at rate λ_1 . Differences in the arrival rates of internal and external promotion offers may reflect, for example, preferences (for or against) hiring CEOs within the firm or differences in firms' cost of internal and external search for executive talent. Upon receiving and accepting an external promotion offer, the executive assumes the CEO role at the poaching firm and their firm-specific capital resets to zero. Lastly, offers for an executive position at a different firm arrive at rate λ_2 , and upon accepting such an offer, the executive switches firms and $k(\tau)$ similarly resets to zero.

For CEOs, the only types of viable outside offers are for CEO positions at different firms; we rule out demotions *ex ante*.¹¹ As in the case of executives, we assume that offers for outside CEO positions arrive at rate λ_1 . We will show that despite this assumption, the rates of executive-to-CEO and CEO-to-CEO transitions will differ in equilibrium. As in all previous cases, CEOs forgo their firm-specific capital $k(\tau)$ upon taking a position with another firm.

In addition to receiving job offers, managers may receive employment shocks inducing separation or exit from the labor market. We assume that managers retire and exit the labor market permanently at rate μ . Alternatively, at rate η manager-firm matches are dissolved and the manager enters unattachment. While unattached, managers may potentially re-enter the labor market by receiving an offer for an executive or CEO position, which arrive at rates γ_E and γ_C , respectively. Note that unattachment is not synonymous with unemployment. Unattached managers have no tie to any publicly-held firm, but may be employed in the government or private

¹¹This assumption is motivated by the data. When examining potential demotions (when a manager is the CEO at a firm in year t and a non-CEO executive in year $t+1$) in our data, we find that a large majority are advisory positions. For example, the previous CEO stays on at the firm explicitly as an advisor to the current CEO, or implicitly by taking chairmanship of the board. However, a small number of true demotions do exist in our data. Our model is not intended to study the decision to promote a (possibly interim) CEO and subsequently demote them, so we remove these managerial spells from our estimation sample.

sector, for example; explicitly modeling these types of employment is beyond the scope of this paper. What is relevant for our purposes is that unattached managers accumulate no firm-specific human capital; their value of k remains at zero.

3.2. Bargaining and Wage Contracts

Compensation takes the form of a piece-rate contract: managers receive an endogenous share $R \in (0, 1]$ of their marginal product $Y(t, \tau, p)$ (Postel-Vinay and Robin, 2002; Bagger et al., 2014). Consequently, log compensation is given by:

$$w(r, t, \tau, p) = r + p + h(t, \tau) \quad (3)$$

Piece rates $r = \log(R) \leq 0$ are determined via strategic bargaining between firms and managers, with $r = 0$ capturing the extreme case in which the manager fully extracts the surplus generated by the position. For a manager with experience t , tenure τ , match productivity p and a contract stipulating piece rate r , we denote the discounted value of their position by $V_i(r, t, \tau, p)$, for $i \in \{E, C\}$.

Throughout their careers, managers leverage competing job offers to garner themselves higher payoffs. Any time an offer arrives, a bargaining game is initiated between the manager and firm(s) bidding for their services. The outcome of the bargaining determines both the manager's (potentially new) position and the piece rate characterizing their new contract. The bargaining protocol is as an extension of Cahuc et al. (2006), which we microfound in full in Appendix A, and the exact form of the bargaining game will depend on the type of job transition in question: internal promotion, external promotion, or lateral move.

3.2.1. Internal Promotions

Consider a manager employed in an executive position with state (r, t, τ, p) , and suppose their firm approaches them with a potential promotion to CEO. First, the manager's match productivity for the CEO role p' is drawn from F , which all parties observe. Given p' , the firm offers piece

rate r' characterizing the manager's contract were they to accept to promotion. The firm's offer satisfies the following condition:

$$\begin{aligned} V_C(r', t, \tau, p') &= V_E(r, t, \tau, p) + \beta(V_C(0, t, \tau, p') - V_E(r, t, \tau, p)) \\ &= \beta V_C(0, t, \tau, p') + (1 - \beta)V_E(r, t, \tau, p) \end{aligned} \quad (4)$$

The value to the manager of accepting promotion, $V_C(r', t, \tau, p')$, is the value of remaining in their current position $V_E(r, t, \tau, p)$ plus a share β of the additional worker rent created by the new position. The parameter β , to be estimated, measures managers' bargaining power. The sharing rule (4) implies that it is in the manager's interest to accept promotion to CEO if and only if $V_C(0, t, \tau, p') > V_E(r, t, \tau, p)$. We define $\bar{\theta}(r, t, \tau, p)$ as the critical level of match productivity satisfying the indifference condition:

$$V_C(0, t, \tau, \bar{\theta}(r, t, \tau, p)) = V_E(r, t, \tau, p) \quad (5)$$

So, an executive with state (r, t, τ, p) accepts an internal promotion only if $p' > \bar{\theta}(r, t, \tau, p)$, upon which they assume the CEO role with piece rate contract satisfying (4). Otherwise, if $p' \leq \bar{\theta}(r, t, \tau, p)$, the executive and firm halt their negotiations and the executive remains in their current position.¹²

3.2.2. Cross-Firm Transitions

When considering an internal promotion, optimal managerial decisions are characterized by a simple threshold rule in which they only accept promotions with sufficiently high values of p' . The simplicity arises because only two parties (the manager and incumbent firm) take part in negotiations. The situation is more nuanced for cross-firm job offers as negotiations entail three parties: the manager, the incumbent firm and the poaching firm. The two firms competitively bid

¹²We assume throughout the paper that managers elect to remain in their current positions in the cases of indifference. Further, we impose the initial condition $\bar{\theta}(r, t, \tau, p_{min}) = p_{min}$, implying that an executive with match productivity p_{min} always accepts an internal promotion upon receiving an offer for one.

for the manager's contract, and the manager exercises bargaining power over both.

Cross-firm bargaining has three possible outcomes. First, if the new position has sufficiently high match productivity p' , the poaching firm ultimately wins the manager. Second, if p' is sufficiently low, the manager ignores the offer entirely. Finally, if p' lies in an intermediate range, the incumbent firm can retain the manager with a renegotiated contract.

Consider a manager employed in a position of type $i \in \{E, C\}$ with state (r, t, τ, p) . Suppose they are offered a position of type $j \in \{E, C\}$ with another firm and assume momentarily that the poaching firm can outbid the incumbent (leading the manager to switch firms). The manager's piece rate r' upon accepting this new position satisfies:

$$V_j(r', t, 0, p') = \beta V_j(0, t, 0, p') + (1 - \beta) V_i(0, t, \tau, p) \quad (6)$$

which again states that the value of accepting the external type- j offer $V_j(r', t, 0, p')$ is a weighted combination of the total surplus of the new match $V_j(0, t, 0, p')$ and the manager's outside option $V_i(0, t, \tau, p)$. The value of accepting the outside offer reflects the fact that the manager will initially have $\tau = 0$, and hence will begin the new job spell with $k = 0$. Additionally, the outside option is no longer the manager's current value of employment, but the value of the best offer they could extract from the losing firm (the incumbent in this case). Appendix A.1 shows that competition between the two firms drives the incumbent to bid the full surplus $V_i(0, t, \tau, p)$ which, while never accepted in equilibrium, is a viable threat point which the manager can use as leverage. Managers are thus in more favorable bargaining positions when external offers arrive (Cahuc et al., 2006).

Such competition may still benefit the manager even if they ultimately elect not to accept the outside position. In particular, the value of rejecting the new offer and retaining employment with the incumbent firm is given by:

$$V_i(r', t, \tau, p) = \beta V_i(0, t, \tau, p) + (1 - \beta) V_j(0, t, 0, p') \quad (7)$$

Similar to the previous case, competition between the two firms drives the losing firm's bid (the

poacher in this case) up to full surplus $V_j(0, t, 0, p')$. Again, while this is not accepted by the manager in equilibrium, the *threat* of acceptance induces the incumbent to favorably renegotiate their contract to encourage the manager to stay.

Importantly, it is possible that a competing offer is not threatening enough to trigger renegotiation by the incumbent. For a type- i manager with state (r, t, τ, p) who receives an outside offer for a position of type j , the incumbent firm is induced to renegotiation only if new match productivity p' exceeds $\underline{\theta}_{ij}(r, t, \tau, p)$, defined by:

$$V_i(r, t, \tau, p) = \beta V_i(0, t, \tau, p) + (1 - \beta) V_j(0, t, 0, \underline{\theta}_{ij}(r, t, \tau, p)) \quad (8)$$

Eq. (8) states that an outside offer with match productivity $\underline{\theta}_{ij}(r, t, \tau, p)$ would leave the manager no better off than their current contract. If $p' < \underline{\theta}_{ij}(r, t, \tau, p)$, the manager ignores the offer. We summarize managers' acceptance criteria in the following theorem:

Theorem 1. *Suppose a type- i manager with state (r, t, τ, p) receives an outside offer for a position of type j . Let p' denote the match productivity associated with this new position. The poacher wins the manager with piece rate r' satisfying (6) only if $p' > x_{ij}$, where:*

$$x_{EC} = \bar{\theta}(0, t, \tau, p) + k(\tau) \quad (9)$$

$$x_{CC} = x_{EE} = p + k(\tau) \quad (10)$$

Additionally, when $r < 0$, we have that:

$$\bar{\theta}(r, t, \tau, p) < \bar{\theta}(0, t, \tau, p) < p \quad (11)$$

The incumbent retains the manager with piece rate r' satisfying (7) only if $p' \in [\underline{\theta}_{ij}, x_{ij}]$. Lastly, if $p' < \underline{\theta}_{ij}$, the offer is ignored.

The new position's match productivity p' determines whether the manager accepts an outside offer, is retained under a renegotiated contract, or ignores to offer entirely. The external

acceptance thresholds (x_{ij}) reflect the fact that in all cases of cross-firm transitions, the match productivity of the new position must be high enough to compensate for the firm-specific capital $k(\tau)$ lost during the job transition. Additionally, the likelihood of accepting an offer declines as $k(\tau)$ increases because firm-specific human capital accumulation gradually increases the opportunity cost of switching firms. This induces a degree of *job lock*: imperfect transferability of human capital reduces the realized rate of cross-firm transitions.

Theorem 1 and inequality (11) provide information about the equilibrium proportions of internally- and externally-hired CEOs. Conditional on receiving an offer p' , the probability of accepting a poaching CEO offer is lower for CEOs than executives.¹³ This is due to different outside options between the two types: CEOs, all else equal, have higher outside options than executives, making them harder to poach.

Second, even without firm-specific capital, internal executives are more likely to accept a CEO offer than outside managers (conditional on an offer arriving). The competitive pressure that arises when attempting to poach outside managers does not affect internal promotions: attempting to poach outside managers can initiate a bidding war and induce the manager's incumbent firm to offer a renegotiated contract and keep the manager. Hence, internal promotions constitute a “path of least resistance” and will be more likely in expectation.

3.2.3. Unattached Managers

Recall that unattached managers can re-enter the labor market with an executive or CEO job offer (which arrive at rates γ_E and γ_C). For tractability, we follow [Bagger et al. \(2014\)](#) and assume that the value of unattachment $V_U(t)$ is equivalent to employment in the least-productive executive position: $V_U(t) = V_E(0, t, 0, p_{min})$. An implication is that unattached managers will accept any offer. Upon receiving an offer of type $j \in \{E, C\}$ with match productivity p' , managers exit unattachment

¹³This arises because outside executives have lower poaching thresholds: for a given $k(\tau)$, $x_{EC} < x_{CC}$ (i.e., $\bar{\theta}(0, t, \tau, p) < p$). For the p -distribution's survivor function $S(\cdot) = 1 - F(\cdot)$, the preceding inequality implies that $S(p + k(\tau)) < S(\bar{\theta}(0, t, \tau, p) + k(\tau))$, so outside CEOs are more likely to stay in their current position if they receive an offer.

and begin their new position with piece rate r' satisfying:

$$V_j(r', t, 0, p') = \beta V_j(0, t, 0, p') + (1 - \beta) V_E(0, t, 0, p_{min}) \quad (12)$$

3.3. Equilibrium Contracts

We next summarize the equilibrium piece rate contracts implied by the bargaining rules outlined above. We focus on piece rates set following transitions into CEO employment; discussion of the piece rates following lateral executive moves is relegated to Appendix A.¹⁴ The specific form of CEOs' contracts will depend on the route that led them to the CEO position. In other words, contracts will be different for CEOs who were promoted from within, promoted from outside, or poached from a CEO position at another firm. To aid exposition, we adopt the following terminology: *horizontal* moves refer to cross-firm CEO to CEO transitions, *diagonal* moves refer to cross-firm executive to CEO transitions, and *vertical* moves refer to within-firm executive to CEO transitions. In general, equilibrium piece rates are of the form $r_i^m(p, \tau_-, z)$ for $i \in \{E, C\}$ and $m \in \{hor, diag, vert\}$, where p is the position match productivity, τ_- is the manager's level of tenure at the time of negotiation, and z is a jump process tracking the manager's outside option.¹⁵

We begin by analyzing the case of horizontal CEO transitions. Consider a CEO employed in a p -productivity position who is poached into a p' position. Let $k_- \equiv k(\tau_-)$ be shorthand for the CEO's level of firm-specific human capital just before leaving their previous firm. The initial piece rate in their new position is given by:

$$r_C^{hor}(p', \tau_-, p + k_-) = -(1 - \beta) \int_{p+k_-}^{p'} q(x) dx \quad (13)$$

¹⁴In Appendix A.2, we first fully derive the executive and CEO value functions, and then use these to derive contracts in closed-firm in Appendix A.3.

¹⁵For example, consider a CEO whose position is characterized by the tuple $(p, k(\tau))$. Suppose they take a CEO position at a new firm. Their outside option z has initial value $p + k(\tau_-)$. Suppose at some point in the future, an outside CEO offer arrives with match productivity p' , which is ultimately rejected. Following this, the CEO's outside option $z = p'$. Thus, outside options jump upon receiving a meaningful offer and remain constant otherwise. The dynamics of the renegotiation boundary $\underline{\theta}_{jk}$ ensure that z is weakly increasing. We suppress the piece rate's dependence on t for notational convenience.

where

$$q(x) = \frac{\rho + \mu + \eta + \lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 S(x)}$$

The piece rate (13) resembles that of [Bagger et al. \(2014\)](#), but differs through its dependence on k_- . A CEO's firm-specific human capital at their previous firm has no effect on output in their new position, yet directly impacts the structure of their new contract. Namely, to compensate managers for firm-specific capital lost upon switching firms, negotiated piece rate increase with respect to their accumulated human capital at their previous firm.

Consider next an executive in a p position who is externally promoted into a p' CEO position. Their initial piece rate is given by:

$$r_C^{diag}(p', 0, p + k_-) = r_C^{hor}(p', 0, p + k_-) - (1 - \beta) \int_{\tilde{\theta}(0, t, \tau_-, p) + k_-}^{p + k_-} q(x) dx \quad (14)$$

Theorem 1 implies that the second term in (14) is negative. This implies that externally-hired executives are initially paid less than externally-hired CEOs (all else equal), which reflects differences in outside options between these two types of managers at the time of the pay negotiation: CEO positions are more valuable sources of employment, making CEOs more expensive to successfully poach than executives.

Finally, consider an executive in a p position who is internally promoted into a p' CEO position. The initial piece rate is given by:

$$\begin{aligned} r_C^{vert}(p', 0, p' + k_-) &= r_C^{diag}(p', 0, p' + k_-) \\ &\quad - (1 - \beta) \left[\int_{p'}^{p' + k_-} q(x) dx + \int_{\tilde{\theta}(r, t, \tau_-, p) + k_-}^{\tilde{\theta}(0, t, \tau_-, p) + k_-} q(x) dx \right] \end{aligned} \quad (15)$$

The above condition highlights the two channels which yield differences in initial pay when comparing internally and externally-promoted CEOs. First, vertically-promoted CEOs retain their firm-specific human capital upon accepting the promotion, so firms do not need to offer internal

hires as much surplus to induce acceptance (reflected in the second term of 15). Additionally, inside executives, unlike outsiders, do not benefit from cross-firm competition when negotiating the terms of a promotion, further decreasing the starting pay of internally-promoted executives (reflected in the third term above).

3.3.1. Renegotiation

We next analyze the effect of outside CEO offers with match productivity $p' \in [\underline{\theta}_{ij}, x_{ij}]$, in which case CEOs are retained and contracts renegotiated. We refer to this interval as the *renegotiation region*, representing the set of competing positions which, while not lucrative enough for the CEO to accept, would provide the CEO with enough leverage over their current employer to induce renegotiation in the event of an offer. Upon receiving an outside offer with productivity p' in this range, the renegotiated piece rate will be:

$$r_C^{reneg}(p, \tau_-, p') = -(1 - \beta) \int_{p'}^{p+k(\tau_-)} q(x) dx \quad (16)$$

Renegotiated piece rates have structure mirroring that of the piece rates sets following employment transitions. As before, they are contingent on incumbent match productivity p , the CEO's tenure at time of negotiations τ_- , and the CEO's outside option which in this case is the match productivity p' of the poaching offer. As discussed previously, the manager will discard offers which do not favorably improve the current contract (i.e., fall below the threshold $\underline{\theta}_{ij}$). The match productivity of the marginal position triggering favorable renegotiation, $\underline{\theta}_{ij}$, evolves with tenure such that the following result holds:

Theorem 2. *For all $ij \in \{CC, EC, EE\}$:*

$$\frac{\partial x_{ij}}{\partial \tau} = \frac{\partial k}{\partial \tau} \geq \frac{\partial \underline{\theta}_{ij}}{\partial \tau} \geq 0 \quad (17)$$

Consequently, the mass of the renegotiation region $[\underline{\theta}_{ij}, x_{ij}]$ grows with tenure for all managers.

Proof. See Appendix A. ■

The above result highlights the effect of firm-specific capital accumulation on managers' capacity to extract surplus through renegotiation. First, as k grows with CEO tenure, the upper bound of the renegotiation region increases. The lower bound of this region, $\underline{\theta}_{jk}$, grows at a rate slower than that of the upper bound, so that an increasing share of offers result in renegotiation. Thus, while firm-specific capital accumulation hampers managerial mobility, it increases their propensity to extract surplus from the incumbent through renegotiation. As managers grow increasingly valuable to their incumbent firms with tenure, firms are increasingly willing to concede favorable renegotiations to the CEO's terms of contract as not to lose them to a competitors. Note that such a phenomenon would be nonexistent in the absence of firm-specific capital accumulation, as $\frac{\partial k}{\partial \tau} = 0$ implies that $\frac{\partial \underline{\theta}_{jk}}{\partial \tau} = 0$, in which case the renegotiation region remains fixed throughout managers' employment spells.

3.4. The Managerial Wage Process

For a manager of type $i \in \{E, C\}$, the equilibrium wage process is given by:

$$w_i(r, t, \tau, p) = a + g(t) + k(\tau) + p + r_i^*(p, \tau_-, z) \quad (18)$$

where r_i^* is the realized piece rate following the manager's most recent pay negotiation. Managerial compensation can be decomposed into a manager fixed effect a , an experience trend $g(t)$, a tenure trend $k(\tau)$, a position fixed effect p , and a random effect r_i^* reflecting the dynamic influence of bargaining, outside options, and cross-firm competition on managerial pay. The empirical value of (18) is apparent, and replicating it forms the basis of our identification strategy. As managers will have faced distinct histories with respect to job search, wage levels and contracts will be path-dependent. Our structural approach allows us to decompose pay into its observable and unobservable primitives, and diagnose the relative importance of each.

4. Estimation

We estimate the model via indirect inference (McFadden, 1989; Smith, 2016). We outline in this section our estimation algorithm, identification strategy, and set of target moments used to recover our structural parameters. We first compute a set of empirical moments using the estimation sample described in Section 2. Then, we simulate the model and compute an identical set of moments using the resulting simulated panel data set. Parameter estimates are selected to minimize the distance between these simulated moments from their empirical counterparts. Further details are included in Appendix B.

4.1. Model solution and simulation

To simulate the model, we start with an initial cross-section of I managers at time 0, drawing each an ability $a \sim N(0, \sigma_a^2)$, and subsequently assigning them to executive and CEO positions based on the rates γ_E and γ_C . Tenure τ and experience t are initialized at zero. For each employed manager, we draw a match productivity p and set the lower renegotiation bound at p_{min} .¹⁶ Following Bagger et al. (2014), we assume that match productivity follows a Weibull distribution:

$$F(p) = 1 - \exp(-s_1(p - p_{min})^{1/s_2}) \quad (19)$$

where s_1 , s_2 , and p_{min} are parameters to be estimated.

Labor market shocks (e.g. separation, job offers, retirement) arrive at the relevant rates for each manager at any given time. In the event of a job offer, a corresponding position productivity p' is drawn and employment outcomes are realized following the discussion in Section 3.3. Managers are immediately replaced in case of retirement (which we force after 45 years of experience if it has not already occurred), so that the stock of managers remains steady over time. We assume that the components of human capital accumulate linearly with experience and tenure so that $g(t) = \delta_g t$ and $k(\tau) = \delta_k \tau$, where we estimate both δ_g and δ_k .

¹⁶All managers are assumed unattached prior to their first position. By (12), their renegotiation region is $[p_{min}, p]$.

Given managers' states, we compute their compensation according to (18). It is worth noting that the set of managerial contracts described in Section 3.3 and subsequent compensation process (18) are convenient as they permit the model to be simulated without numerically computing managers' value functions. We simulate the model for a total of 80 years, using the first 50 years as a "burn-in" period to allow the economy to approach a steady-state.¹⁷ After the burn-in period, we are left with a 30-year panel dataset to compute simulated moments.

4.2. Identification Strategy

We show there is a tight relation between the reduced-form moments of the auxiliary models and structural parameters. Our three sets of moments target different sets of structural parameters, and our identification argument follows key papers in the structural search and human capital literature (e.g., Cahuc et al., 2006; Bagger et al., 2014); crucial to its success is separating the impacts of search frictions, imperfect labor market competition, and firm-specific human capital accumulation in determining compensation and mobility over the executive's career.

4.2.1. Managerial Labor Market Mobility

Over their careers, managers transition across three labor market states: unattachment, executive employment, and CEO employment. CEO spells may end with unattachment, retirement, or a horizontal move. Executive spells with an internal promotion, external promotion, or horizontal move. Unattached managers may re-enter the labor market through offers of executive or CEO employment. To identify the arrival rates corresponding to each of these labor market transitions, we fit Kaplan-Meier estimates of the survivor function for each type of employment and transition event. As an example, consider CEO-to-CEO transitions. Let $N_{CC}(\tau)$ be number of CEOs at tenure τ who can experience a CEO-to-CEO transition and let $M_{CC}(\tau)$ be the number of CEOs that do

¹⁷As an alternative, we could in principle derive the relevant steady-state distributions analytically and initialize individual states by taking draws from these distributions. Unfortunately, obtaining an analytic characterization of the steady state is intractable in our case. We therefore follow previous literature (e.g., Bobba et al., 2021) and take the approach of obtaining model stationarity via forward simulation.

actually take up CEO employment. The Kaplan-Meier estimate of the survival function is

$$\hat{S}_{CC}(\tau) = \prod_{s=0}^{\tau} \frac{N_{CC}(s) - M_{CC}(s)}{N_{CC}(s)}.$$

the survival function estimates are defined analogously for all other labor market events. Executive and CEO survival functions are defined over tenure, while unattachment are defined over experience (as tenure does not accumulate for unattached managers). In each case, we target the quantities $1 - \hat{S}(5)$, $\hat{S}(5) - \hat{S}(10)$, $\hat{S}(10) - \hat{S}(15)$ and $\hat{S}(15) - \hat{S}(25)$, which give the estimated probability that a manager experiences an event within each respective time frame.

4.2.2. Mincer Wage Regressions

Our model admits a closed-form, Mincer-like process (18) for managerial compensation as a function of experience, tenure, ability, position productivity, and labor market competition. As a consequence, it is natural to target the coefficients of the analogous Mincerian wage regression Mincer (1974). In particular, for firm i , position j , manager m and year y :

$$w_{ijmy} = \sum_{q=1}^3 (a_q \tau_{my}^q + b_q t_{my}^q) + \delta_{\text{prom}} \text{Prom}_{ijmy} + \delta_{\text{move}} \text{Move}_{ijmy} + \gamma_{ij} + \phi_m + u_{ijmy}, \quad (20)$$

τ^q and t^q represent tenure and experience along with their second and third-order polynomial terms. In the model, as long as the manager remains in the workforce, experience accumulates, whereas tenure resets to zero each time a manager joins a new firm. Conditional on the position, manager, and labor market history, variation in wages across differing levels of experience and tenure help identify the trends $g(t)$ and $k(\tau)$.

The firm $\times$ position (CEO vs. non-CEO) fixed effect γ_{ij} helps identify the parameters of the position productivity distribution (19). We winsorize γ_{ij} at the 0.05 percentile and include the first four moments of the distribution as moments to target. The volatility of the manager-specific fixed effect identifies variation in managerial ability.

The coefficients δ_{prom} and δ_{move} are informative of the bargaining power parameter β . Note

that managers' piece rate will be a function of the history of job offers they receive. That is, for manager m , let $N_m(t)$ denote the number of job offers m has received up until experience t that led to a revision of the piece-rate:

$$r_m(N_m(t)) = r_m(0) + \sum_{k=1}^{N_m(t)} \underbrace{r_m(k) - r_m(k-1)}_{\delta_{km}(\beta)},$$

where $\delta_{km}(\beta)$ captures the effect of offer k on manager m 's contract, which ultimately depends on their bargaining power β . While job offers that trigger renegotiation are unobserved in the data, internal promotions and external job transition events will update manager m 's contract, so including indicators that switch in the year that a manager is internally-promoted or first experienced a job-to-job transition aid the identification of β .

4.2.3. Within- and Across-Job Wage Growth

We separately target the impacts of experience and tenure on wage growth within- and across-jobs in our wage growth regression, which proves crucial for separately identifying the impacts general and firm-specific human capital accumulation.

We focus on managers with at least two years of consecutive experience as a manager, and estimate different tenure and experience polynomial trends for managers that remain employed at the same firm (stayers) against those that switch firms (movers).

To isolate how wages change for movers, we define the indicator M_{my} which tracks the 3-year window after a job-switch event for manager m , allowing us to track wage changes when managers switch firms. We then estimate, for firm i , position j , manager m and year y

$$\Delta w_{ijmy} = \overbrace{\sum_{q=2}^3 (c_q \Delta t_{my}^q + d_q \Delta \tau_{my}^q)}^{\text{Stayers}} + \underbrace{\sum_{q=2}^3 (c_{Mq} [\Delta t_{my}^q \times M_{my}] + d_{Mq} [\Delta \tau_{my}^q \times M_{my}])}_{\text{Movers}} + \gamma_{ij} + \Delta u_{ijmy} \quad (21)$$

For movers, the coefficients c_{Mq} and d_{Mq} are important for separately identifying k and g as they describe the impact of variation in a manager’s experience and tenure when they switch firms on wage growth. Eq. (21) also provides a statistical description of within-job wage growth by separately modeling wage growth for stayers, which help pin down the estimated trends in human capital accumulation. Finally, we also include position $\times$ firm fixed effects, which nets out how differences in position impact wage growth.

It is important to note that (20) and (21) do not have a structural interpretation (Bagger et al., 2014). Indeed, our model would state they are misspecified and suffer from omitted variable bias as we do not observe the full impact of labor market competition on managerial wages. Our identification strategy aims to replicate the misspecification and use the structural model to uncover the relevant omitted variables in these regressions.

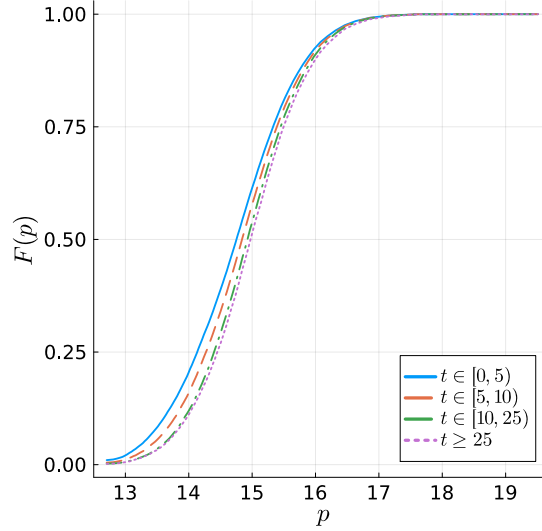
5. Structural Parameter Estimates

Parameter estimates are reported in Table 2. Regarding transitions into CEO employment, our estimates of λ_0 (.156) and λ_1 (.084) imply that internal CEO offers arrive at roughly twice the rate of external offers. There are several factors which may explain this disparity: the prevalence of non-compete agreements (Shi, 2023), external search costs (Geelen and Hajda, 2024), or non-pecuniary preferences for internal candidates (He and Schroth, 2024; Capron et al., 2024). While this gap is considerable, the gap in internal and external CEO transitions is amplified by limitations in human capital portability.

Our estimates of δ_g (.032) and δ_k (.036) imply that general and firm-specific human capital accumulate at roughly the same rate. The trajectory of total human capital $h(t, \tau)$, however, is contingent on managers’ career path. As an example, consider two managers of equal ability (a), both with 10 years of experience. Total human capital $h(t, \tau)$ will be 18% higher for the manager who stays in the same position relative to one who switched positions in 5th year of their career. For managers, enhanced human capital accumulation due to job stability raises the opportunity cost of switching firms and contributes in part to the empirically low rate of cross-firm mobility.

Figure 1. Position productivity distribution

This figure displays the estimated position productivity distribution, which is characterized by location p_{min} , shape s_1 and scale s_2 from Table 2. In the figure, we display the (simulated) empirical CDFs of position-types, conditional on experience t .

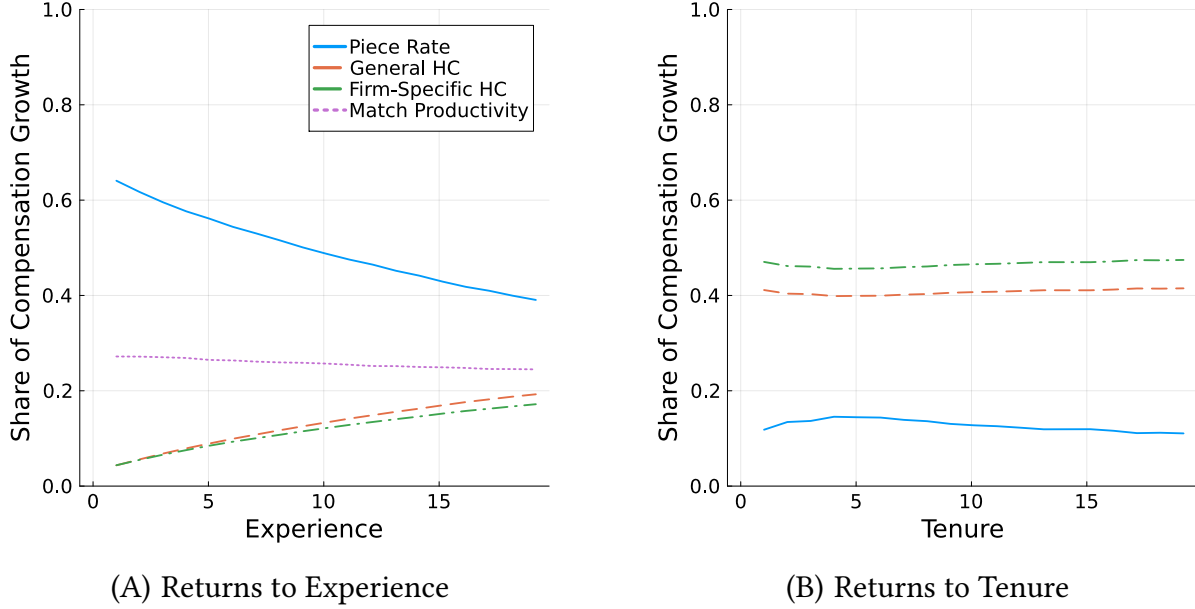


Our estimates of the distributional parameters s_1 , s_2 , and p_{min} imply that heterogeneity in match productivity (p) is substantial; increasing match productivity from its mean (13.0) by one standard deviation is associated with an 22% increase in total compensation. Further, while we do not impose any direct dependence of p on experience, the equilibrium distributions of match productivity vary systematically with experience. We illustrate this in Figure 1, which plots the cumulative distributions of simulated match productivity separately different levels of managerial experience. In terms of stochastic dominance, match productivity generally increases with experience. Hence, though the market is rigid, managers have some capacity to transition into higher-quality matches over their careers. Endogenous sorting of this nature has important empirical implications. As experience is correlated with unobservable match productivity, naive regressions of managerial compensation on experience will yield biased coefficient estimates. To decouple the compensation effects of sorting and human capital accumulation, we decompose CEO compensation growth over experience and tenure in Figure 2.

In Figure 2 Panel A, we first compute average CEO compensation growth for each year of experience. We then compute average growth of each experience-varying component of CEO

Figure 2. CEO Compensation Growth Decomposition

This figure accumulated wage growth for CEOs over tenure and experience. We use (18) to decompose accumulated wage growth into changes in the piece rate, general and firm-specific human capital, and position productivity (which does not play a role for tenure).



pay: general human capital, firm-specific human capital, piece rates, and match productivity. Finally, to compute the share of compensation growth attributable to growth in each component, we scale the growth of each component by total compensation growth. Figure 2 Panel B displays the same decomposition within-position (we omit match productivity as it is fixed within a given position). When explaining the returns to experience, Panel A suggests that human capital (both general and firm specific) plays a fairly minor role. Increases in the CEO shares of rents, measured here by the piece rate, explain the largest share of compensation growth over experience, particularly in early stages of careers. In addition, the contribution of changes match productivity is substantial, explaining a stable 30% of total compensation growth over experience.

The returns to tenure, (Panel B of Figure 2), tell a different story. Unlike the previous case, human capital is the main driver of within-position compensation growth, with general human capital being slightly more important. The contribution of piece rate growth is modest and remains stable throughout CEO tenure. The stability of the within-position piece rate contribution suggests that in spite of the job-lock effect, the influence of outside competition on CEO compen-

sation (i.e. renegotiation) is persistent. However, comparing this with the piece rate contribution in the returns to experience, it is clear that within-position piece rate renegotiations are less consequential than cross-position piece rate increases due to job transitions.

6. CEO Bargaining Power and Labor Market Competition

Our estimate of β (.136) implies a modest level of managerial bargaining power, lower than previous estimates in the literature.¹⁸ In our model, managers capture rents through both bargaining power and by leveraging cross-firm competition to improve contracts: bargaining power and competition work in tandem to increase managerial compensation. By explicitly modeling the influence of outside competition on managerial pay-setting, we can decouple the effects of competition from pure bargaining power in determining CEOs' surplus capture (Cahuc et al., 2006).

This is particularly important in the context of corporate executives. The high observed share of managerial rents is often taken as evidence of managers exercising undue influence on the pay-setting process (Bebchuk and Fried, 2006). Our results suggest that surplus capture may instead be the product of well-functioning labor market competition. In this section, we present a decomposition of managers' observed share of rents into components attributable to managerial bargaining and labor market competition. We then provide an assessment of the bias that results from attempting to estimate CEO bargaining power in the absence of an explicitly-modeled labor market.

6.1. Decomposition of CEO Surplus

Our model provides a simple way to separate the compensation effects of bargaining power and competition. Following Cahuc et al. (2006), we derive a “naive” estimate of bargaining power in an alternative version of the model with no on-the-job search, and hence no cross-firm competition. Comparing this alternative estimate of bargaining power to our structural estimate from the main model then gives the share of managers' observed surplus explainable by competitive forces as opposed to pure bargaining power.

¹⁸For example, Taylor (2013) estimates that CEOs capture about 50% of surplus given upside news about their ability.

This experiment amounts to setting CEOs' on-the-job search parameter λ_1 to zero. This forces the CEO's outside option to unattachment, which in turn ascribes any CEO surplus in excess of unattachment value to pure bargaining power.¹⁹ Consider for example a CEO with match productivity p , tenure τ , and experience t . With no on-the-job search, the log wage equation must be of the form:

$$w(p, t, \tau) = b(p, \tau, \underline{\theta}_C)(p + k(\tau)) + (1 - b(p, \tau, \underline{\theta}_C))p_{min} + a + g(t), \quad (22)$$

where b measures the CEO's observed share of match surplus. Hence, in the absence of on-the-job search, $\mathbb{E}[b]$ would be our estimate of CEO bargaining power. We can relate this to bargaining power in the main model (β) by setting equal the structural wage equation and the counterfactual wage equation (22):

$$\begin{aligned} b(p, \tau, \underline{\theta}_C) &= \frac{r(p, \tau, \underline{\theta}_C) + p + k(\tau) - p_{min}}{p + k(\tau) - p_{min}} \\ b(p, \tau, \underline{\theta}_C) &= 1 + \frac{r(p, \tau, \underline{\theta}_C)}{p + k(\tau) - p_{min}} \end{aligned} \quad (23)$$

Note that $b(p, \tau, \underline{\theta}_C) \in [\beta_1, 1]$,²⁰ so the bias that arises when neglecting competition is weakly positive. We can then decompose average total surplus b into a bargaining component and competition component using the simple identity:

$$1 = \underbrace{\mathbb{E}\left[\frac{\beta}{b(p, \tau, \underline{\theta}_C)}\right]}_{\text{CEO bargaining power}} + \underbrace{\mathbb{E}\left[\frac{b(p, \tau, \underline{\theta}_C) - \beta}{b(p, \tau, \underline{\theta}_C)}\right]}_{\text{labor market competition}} \quad (24)$$

With this decomposition, we estimate $b(p, \tau, \underline{\theta}_C)$ in our simulated sample and report results in

¹⁹In our model, unattachment has the same value as an offer from the least productive firm, p_{min} .

²⁰If $r(p, \tau, \underline{\theta}_C) = 0$, then $b = 1$. The minimum value of $r(p, \tau, \underline{\theta}_C)$ arises when the true λ_1 equals zero, where

$$r(p, \tau, p_{min}) = -(1 - \beta_1)(p + k(\tau) - p_{min})$$

and $b = \beta$.

Table 4. In the full sample, our naive estimate of CEO bargaining power is roughly .564, in line with previous estimates in the literature. Comparing this to our estimate of CEO bargaining power β , it is clear that a large proportion of CEO surplus capture is driven by competition. We find that roughly 75.9% of the observed share of CEO surplus is the product of labor market competition as opposed to pure bargaining power. Despite the low empirical rate of CEO cross-firm mobility, the influence of competition on CEO pay is substantial.

Additionally, we report a number of subsample estimates of $b(p, \tau, \theta_C)$ to illustrate that this bias varies systematically across observables. Panel A of Table 4 reports estimates separately across CEO types, namely whether a given CEO was promoted from within, poached from a CEO position, or poached from an executive position. Panels B and C report estimates across the distributions of experience and tenure, respectively. Starting with the CEO type subsamples, we again see upward bias in the bargaining power estimate in all cases. The magnitude of this bias, however, varies across CEO type. The bias is largest among CEOs who are poached from competing firms, whether from a CEO or executive position. Attempting to poach a manager initiates a bidding war with the incumbent, introducing a degree of competition which the manager can then leverage to negotiate a more favorable contract. This is distinct from the internal bargaining regime, in which there is no direct influence of outside competition. The surplus capture of poached managers is thus more attributable to competition.

Further, we see heterogeneous effects of labor market competition on CEOs' captured share of surplus across both the distribution of experience and tenure. Panel B of Table 4 shows that estimates of $b(p, \tau, \theta_C)$ increase with managerial experience. Put differently, the share of CEO surplus explained by labor market competition increases with experience. This follows from the history-dependence of piece-rate contracts. The impact of a one-off bidding war has a persistent effect on managers' compensation throughout their careers. As experience increases, these extracted benefits accumulate and comprise an increasing share of their total captured surplus.

Similarly, competition explains an increasing share of CEO surplus as tenure progresses, and the rationale is similar to the previous result. As tenure increases, CEOs receive a growing number

of outside offers which trigger renegotiations with their incumbent employers. This gradually increases CEOs' share of total surplus, despite their bargaining power remaining constant.

7. Firm-Specific Human Capital, Mobility, and CEO Rents

The previous section highlights the large influence of competition on CEO pay. This influence may be somewhat surprising: the low-degree of mobility in the managerial labor market may suggest that competition plays a small role. If the overwhelming majority of CEOs work for only one firm over their career, how can cross-firm competition have such an impact on compensation? This apparent puzzle is the focal point of this section, which we resolve by illustrating the impact of firm-specific human capital on both CEO cross-firm mobility and CEOs' ability to capture rents.

To do so, we compare CEO compensation and employment outcomes in the estimated model to a counterfactual version of the model which eliminates firm-specific human capital. We first discuss how much firm-specific human capital impacts mobility in the CEO labor market. We then examine how managerial bargaining outcomes, and the persistent effect of competition, are affected by their human capital. Though firm-specific human capital hampers managerial mobility, we find that it enhances their capacity for rent extraction.

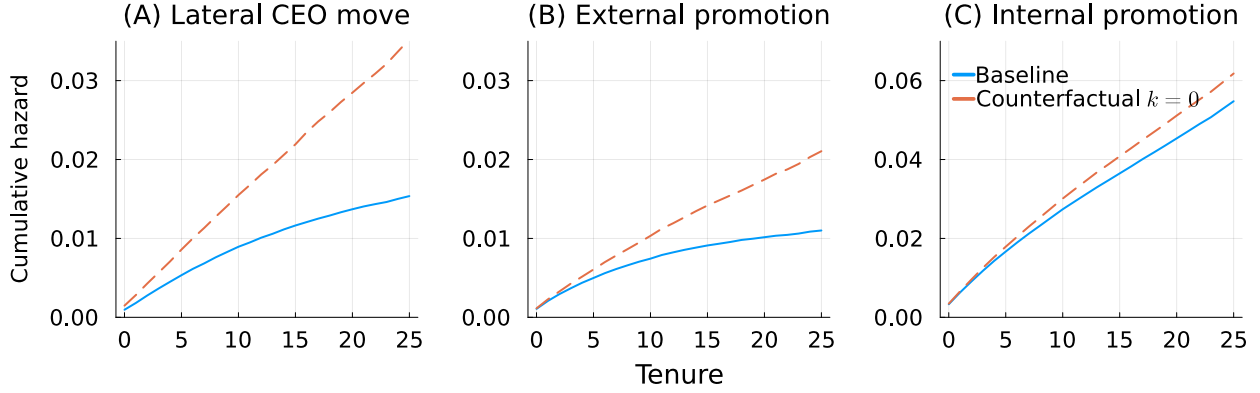
7.1. Human Capital and Transitions into CEO Employment

We first discuss the effect of firm-specific human capital accumulation on cross-firm mobility in the managerial labor market. In Figure 3, we compare the baseline and counterfactual probabilities of transitioning into CEO employment through a lateral move, external promotion, or internal promotion.

When human capital has a firm-specific component, there is a marked decline in the rate of cross-firm transitions, whether a lateral CEO move or external promotion. This results from job lock: as firm-specific human capital accumulates, the opportunity cost of switching employers gradually increases. However, even in the absence of firm-specific human capital, the rate of cross-firm transitions is still quite low. For example, the counterfactual rate of lateral CEO moves

Figure 3. Firm-specific human capital and transition rates

This figure displays cumulative hazard rates of a manager experiencing (A) a lateral CEO move (conditional on being a CEO); (B) an external promotion (conditional on being an executive); and (C) an internal promotion (conditional on being an executive). The blue line displays the estimated rates, the orange lines the counterfactual when $k = 0$.



after 20 years of tenure is only about 3%.

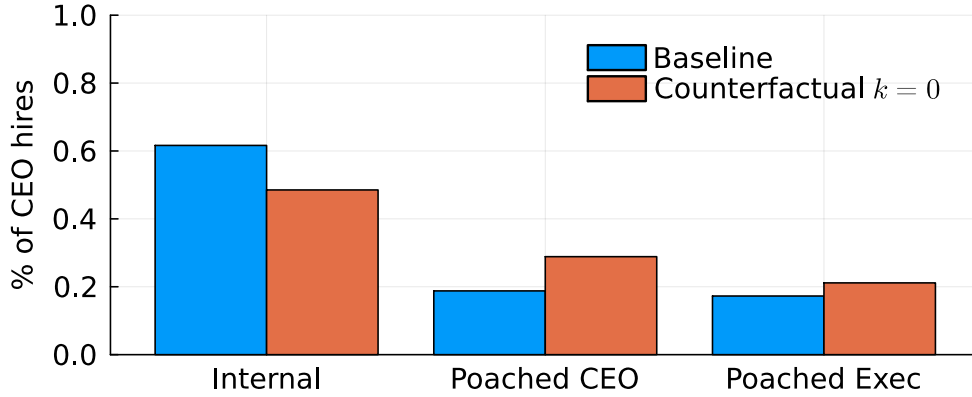
In contrast with cross-firm transition rates, the rate of internal promotions is virtually unaffected by firm-specific human capital. This may come at no surprise, as managerial capital is fully retained upon switching positions within a given firm. Though this rate is stable, the increased frequency of cross-firm transitions will imply that a smaller share of CEOs are appointed from within the firm when firm-specific human capital is immaterial.

We illustrate this in Figure 4, which plots the baseline and counterfactual shares of CEOs hired from within the firm, poached from CEO positions, or poached from executive positions. As expected, the proportion of internally-hired CEOs decreases when eliminating firm-specific capital while the share of external hires increases. Clearly, internal promotions remain the most common route to CEO employment even in the counterfactual case. Thus, while firm-specific human capital accumulation explains a portion of firms’ “preference” for hiring internally, other factors also contribute to this hiring disparity.

The remaining gap in rates of CEO appointments is attributable to a combination of exogenous and endogenous characteristics of the executive labor market. In our model, the key exogenous explanatory features are the different arrival rates of job offers across managerial types. As previously discussed, we estimate these rates to be different ($\lambda_0 > \lambda_1$), implying that insiders are significantly more likely to receive offers in the first place. As these parameters are exogenously

Figure 4. Firm-specific human capital and internal vs. external CEO hiring

This figure displays baseline and counterfactual (with $k = 0$) proportions of internal CEO promotions, externally poached CEOs and externally poached non-CEO executives.



given in our model, whether this gap in arrival rates is the product of inefficient preference for internal candidates or the efficient bypassing of external search costs is unclear, and outside the scope of our analysis. However, our model highlights that the disparity in hiring rates is due in large part to endogenous labor market competition.

When attempting to poach a manager, firms must contend with the manager's incumbent employer. This decreases the likelihood of a successful poach, as incumbent employers have the capacity to dissuade managerial exit via contract renegotiations, and negotiations with internal candidates are free from such competitive pressure and more likely to result in a successful CEO appointment. Hence, the high share of internal CEO hires is not entirely the byproduct of labor market frictions, an outcome to be expected in a well-functioning labor market.

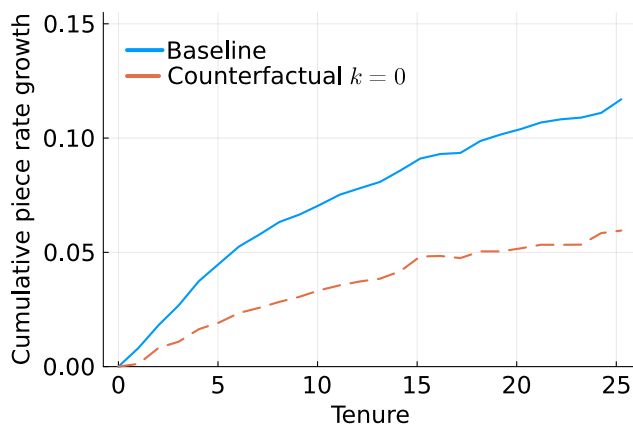
7.2. Firm-Specific Human Capital and CEO Rent Extraction

The previous discussion highlights the following result: the lack of cross-firm CEO movement is in part the consequence of cross-firm competition. Firms' defensive bidding tactics simultaneously decrease the rate of CEO transitions and increase the level of CEO pay. These bidding tactics take the form of renegotiations in our model, and are the key channel through which CEOs can dynamically increase their share of rents within a given position.

As shown in Section 3, conditional on receiving a competing job offer, the likelihood of con-

Figure 5. Firm-specific human capital and CEO rent extraction

This figure displays how the CEO’s accumulated piece growth changes over tenure, both in the main simulated model and in a counterfactual where firm-specific human capital is zero.



tract renegotiations is strictly higher with firm-specific human capital. This arises as firm-specific capital raises the match-specific quality between the firm and CEO, thereby decreasing the likelihood that poachers can outbid the incumbent. Rather than switch positions, CEOs elect to remain with their employers under a renegotiated contract. In Figure 5, we compare average within-position piece rate growth in the baseline and counterfactual versions of our model.

Firm-specific human capital roughly doubles the rate at which CEO piece rates grow with tenure. This is because renegotiations become more likely than transitions when human capital does not transfer perfectly across firms. Though firm-specific capital dampens mobility (Figure 3), it magnifies the positive impact of competition on within-firm compensation growth. By raising the opportunity cost of losing the CEO, it raises the incumbent’s likelihood of matching attractive outside offers, enhancing the CEO’s capacity to extract surplus from the firm.

8. Conclusion

We present a structural model of the managerial labor market that quantifies the relative importance of general and firm-specific human capital accumulation, managerial bargaining power, and labor market competition in determining compensation and mobility. Our model allows for internal and external promotions and admits closed-form expressions for equilibrium managerial contracts. We theoretically show that realized CEO surplus capture is path-dependent: poached

CEOs capture more rents than poached non-CEO executives and internally promoted CEOs.

Our paper makes two primary empirical contributions. First, we measure the relative importance of general and firm-specific human capital in managerial skill over the manager's career, and show that firm-specific human capital lowers cross-firm mobility (less job-switching), but raises within-firm mobility (higher surplus capture with current firm). Second, we decompose realized CEO surplus capture into contributions from pure CEO bargaining power and labor market competition and show that labor market competition is generally the larger component.

Our work has important implications for the corporate finance literature. First, our decomposition of managerial human capital is crucial for understanding the CEO labor market, particularly the literature focusing on general CEO skill ([Gabaix and Landier, 2008](#); [Murphy and Zabojnik, 2007](#); [Cziraki and Jenter, 2024](#); [Graham et al., 2020](#)). To the best of our knowledge, our paper is the first in the literature to quantify directly the composition of managerial human capital and the impact of firm-specific skill on managerial wages.

Second, our decomposition of realized CEO surplus is important for understanding the role of CEO bargaining power and agency frictions in determining CEO compensation. We show that, even in an imperfect labor market with search frictions, in which CEO-CEO transitions are rare, CEO surplus capture can largely be explained by competition in the labor market. In models without an explicit managerial labor market, estimates of agency frictions which induce high CEO wages or low turnover may be over-stated.

References

- Bagger, J., Fontaine, F., Postel-Vinay, F., and Robin, J.-M. (2014). Tenure, experience, human capital, and wages: A tractable equilibrium search model of wage dynamics. *The American Economic Review*, 104(6):1551–1596.
- Barry, J. W. (2023). Shareholder voice and executive compensation. *Working Paper*.
- Bebchuk, L. A., Cremers, K. M., and Peyer, U. C. (2011). The ceo pay slice. *Journal of financial Economics*, 102(1):199–221.
- Bebchuk, L. A. and Fried, J. M. (2003). Executive compensation as an agency problem. *Journal of economic perspectives*, 17(3):71–92.
- Bebchuk, L. A. and Fried, J. M. (2006). Pay without performance: Overview of the issues. *Academy of Management Perspectives*, 20(1):5–24.
- Bertrand, M. and Mullainathan, S. (2001). Are ceos rewarded for luck? the ones without principals are. *The Quarterly Journal of Economics*, 116(3):901–932.
- Bobba, M., Flabbi, L., Levy, S., and Tejada, M. (2021). Labor market search, informality, and on-the-job human capital accumulation. *Journal of Econometrics*, 223(2):433–453. Annals issue: Implementation of Structural Dynamic Models.
- Cahuc, P., Postel-Vinay, F., and Robin, J.-M. (2006). Wage bargaining with on-the-job search: Theory and evidence. *Econometrica*, 74(2):323–364.
- Capron, A., Hamilton, B., Hincapié, A., and Lyman, N. (2024). Insider ceos: Lucky or good? *Working Paper*.
- Coles, J. L., Daniel, N. D., and Naveen, L. (2014). Co-opted boards. *The Review of Financial Studies*, 27(6):1751–1796.
- Core, J. E., Holthausen, R. W., and Larcker, D. F. (1999). Corporate governance, chief executive officer compensation, and firm performance. *Journal of financial economics*, 51(3):371–406.
- Custódio, C., Ferreira, M. A., and Matos, P. (2013). Generalists versus specialists: Lifetime work experience and chief executive officer pay. *Journal of Financial Economics*, 108(2):471–492.
- Cziraki, P. and Jenter, D. (2024). The market for ceos. *Working Paper*.
- Duffie, D. and Singleton, K. J. (1993). Simulated moments estimation of markov models of asset prices. *Econometrica*, 61(4):929–952.
- Edmans, A., Gabaix, X., and Landier, A. (2009). A multiplicative model of optimal ceo incentives in market equilibrium. *The Review of Financial Studies*, 22(12):4881–4917.
- Fahlenbrach, R. (2009). Shareholder rights, boards, and ceo compensation. *Review of Finance*, 13(1):81–113.

- Frydman, C. (2019). Rising through the ranks: The evolution of the market for corporate executives, 1936–2003. *Management Science*, 65(11):4951–4979.
- Gabaix, X. and Landier, A. (2008). Why has ceo pay increased so much? *The quarterly journal of economics*, 123(1):49–100.
- Gao, J., Wang, W., and Wu, Y. (2021). Human capital portability and worker career choices: Evidence from m&a bankers. *Available at SSRN 3724657*.
- Gayle, G.-L., Golan, L., and Miller, R. A. (2015). Promotion, turnover, and compensation in the executive labor market. *Econometrica*, 83(6):2293–2369.
- Geelen, T. and Hajda, J. (2024). Succession. *Working paper*.
- Graham, J. R., Grennan, J., Harvey, C. R., and Rajgopal, S. (2022). Corporate culture: Evidence from the field. *Journal of financial economics*, 146(2):552–593.
- Graham, J. R., Kim, D., and Kim, H. (2020). Executive mobility in the united states, 1920 to 2011. *Working Paper*.
- He, L. and Schroth, E. J. (2024). Why are ceos rarely succeeded by external candidates? *Available at SSRN 4728228*.
- Jarosch, G. (2023). Searching for job security and the consequences of job loss. *Econometrica*, 91(3):903–942.
- Lazear, E. P. (2009). Firm-specific human capital: A skill-weights approach. *Journal of political economy*, 117(5):914–940.
- Lyman, N. (2024). Reputation, entrenchment, and dynamic managerial incentives. *Working Paper*.
- Ma, S., Wang, W., and Wu, Y. (2023). Steering labor mobility through innovation. *Working Paper*.
- McFadden, D. (1989). A method of simulated moments for estimation of discrete response models without numerical integration. *Econometrica: Journal of the Econometric Society*, pages 995–1026.
- Mincer, J. (1974). Schooling, experience, and earnings. *Human Behavior & Social Institutions* No. 2.
- Morse, A., Nanda, V., and Seru, A. (2011). Are incentive contracts rigged by powerful ceos? *The Journal of Finance*, 66(5):1779–1821.
- Murphy, K. J. and Zabojnik, J. (2004). Ceo pay and appointments: A market-based explanation for recent trends. *American economic review*, 94(2):192–196.
- Murphy, K. J. and Zabojnik, J. (2007). Managerial capital and the market for ceos. *Available at SSRN 984376*.
- Page, T. B. (2018). Ceo attributes, compensation, and firm value: Evidence from a structural estimation. *Journal of Financial Economics*, 128(2):378–401.

- Postel-Vinay, F. and Robin, J.-M. (2002). Equilibrium wage dispersion with worker and employer heterogeneity. *Econometrica*, 70(6):2295–2350.
- Rubinstein, A. (1982). Perfect equilibrium in a bargaining model. *Econometrica*, pages 97–109.
- Shi, L. (2023). Optimal regulation of noncompete contracts. *Econometrica*, 91(2):425–463.
- Smith, A. A. (2016). *Indirect Inference*, pages 1–6. Palgrave Macmillan UK, London.
- Taylor, L. A. (2010). Why are ceos rarely fired? Evidence from structural estimation. *The Journal of Finance*, 65(6):2051–2087.
- Taylor, L. A. (2013). Ceo wage dynamics: Estimates from a learning model. *Journal of Financial Economics*, 108(1):79–98.
- Terviö, M. (2008). The difference that ceos make: An assignment model approach. *American Economic Review*, 98(3):642–668.

Table 1. Summary and labor market transition statistics

This table displays summary statistics for managerial wages, and labor market experience and tenure (Panel A); and labor market transition statistics (Panel B). In Panel A, statistics are displayed based on employment status (executive or CEO). In Panel B, labor market transition statistics are displayed conditional on employment state (executive, CEO, unattached); the first two columns display the number of transitions and the proportion of observations conditional on state. Columns 3 and 4 display the average years of experience or tenure that the transition occurs, respectively.

Panel A: Summary statistics						
	N	Mean	Std Dev	25%	50%	75%
CEOs						
Log wage	60,104	14.859	1.209	14.071	14.919	15.687
Wage (\$ million)	60,104	5.351	10.943	1.291	3.016	6.497
Wage Growth (%)	55,188	10.420	62.775	-14.997	7.436	35.380
Experience	60,104	22.105	11.583	13	21	31
Tenure	60,104	15.251	11.368	6	12	22
Non-CEO executives						
Log wage	263,729	13.791	1.070	13.051	13.773	14.508
Wage (\$ million)	263,729	1.748	3.915	0.466	0.959	1.999
Wage Growth (%)	211,378	13.457	58.116	-11.005	9.390	36.543
Experience	263,729	15.251	11.916	5	12	23
Tenure	263,729	5.772	5.865	2	4	7

Panel B: Labor market transition statistics				
	N	%	Experience	Tenure
CEOs				
Horizontal CEO move	246	0.409%	21.626	12.098
Unattachment	530	0.882%	20.517	11.485
Retirement	8,687	14.453%	23.607	15.521
Non-CEO executives				
Internal CEO promotion	4,568	1.732%	16.914	9.959
External CEO promotion	329	0.125%	16.581	5.854
Horizontal executive move	1,574	0.597%	14.386	4.810
Unattachment	5,109	1.937%	15.062	4.610
Retirement	41,215	15.628%	15.689	6.343
Unattached managers				
Executive position	5,208	23.216%	19.253	
CEO position	698	3.111%	22.232	

Table 2. Structural parameter estimates

This table displays the estimated structural parameters, given the estimation sample describe in Table 1.

μ	η	λ_0	λ_1	λ_2	γ_E	γ_C	β	σ_a	p_{min}	s_1	s_2	δ_g	δ_k
0.038	0.040	0.156	0.084	0.009	0.132	0.988	0.136	0.439	12.7	1.88	1.57	0.032	0.036

Table 3. CEO human capital composition across experience and tenure

This table shows a decomposition of accumulated CEO human capital into general human capital $g(t)$, and firm-specific human capital $k(\tau)$. Panel A displays the decomposition across labor market experience t and Panel B does so for firm tenure. We compute shares in levels: $s_x = \frac{\exp(x)}{\exp(g)+\exp(k)}$ for each $x \in \{g, k\}$.

Panel A: Human capital decomposition by labor market experience

	(1) Full sample	(2) Conditional on experience (t)	(3)	(4)	(5)
		[0, 5)	[5, 10)	[10, 25)	≥ 25
General human capital (g)	54.36%	50.14%	51.19%	54.16%	59.43%
Firm-specific human capital (k)	45.64%	49.86%	48.81%	45.84%	40.57%

Panel B: Human capital decomposition by firm tenure

	(1) Full sample	(2) Conditional on tenure (τ)	(3)	(4)	(5)
		[0, 5)	[5, 10)	[10, 25)	≥ 25
General human capital (g)	54.36%	55.22%	54.96%	53.93%	50.36%
Firm-specific human capital (k)	45.64%	44.78%	45.04%	46.07%	49.64%

Table 4. CEO surplus capture decomposition

This table shows realized CEO surplus capture. Panel A displays realized CEO surplus capture by eventual CEO hire type. Panel B displays across labor market experience and Panel B by firm tenure. We display realized CEO surplus capture $\mathbb{E}[b \mid X]$, for comparison against exogenously estimated pure CEO bargaining power $\beta_1 = 13.6\%$ in (24). The first row displays average realized surplus capture b . The second row displays the share attributable to labor market competition

Panel A: CEO surplus capture decomposition by CEO hire type

	(1) Full sample	(2) Conditional on CEO hire type	(3) Internal	(4) Poached CEO	(5) Poached exec
Realized surplus capture (b)	56.35%		51.54%	68.64%	64.22%
Labor market competition ($\frac{b-\beta_1}{b}$)	75.94%		73.70%	80.25%	78.89%

Panel B: CEO surplus capture decomposition by labor market experience

	(1) Full sample	(2) Conditional on experience (t)	(3) [0, 5)	(4) [5, 10)	(5) [10, 25)	(6) ≥ 25
Realized surplus capture (b)	56.35%		24.23%	52.57%	63.59%	65.45%
Labor market competition ($\frac{b-\beta_1}{b}$)	75.94%		44.05%	74.21%	78.68%	79.29%

Panel C: CEO surplus capture decomposition by firm tenure

	(1) Full sample	(2) Conditional on tenure (τ)	(3) [0, 5)	(4) [5, 10)	(5) [10, 25)	(6) ≥ 25
Realized surplus capture (b)	56.35%		42.47%	55.64%	61.66%	61.59%
Labor market competition ($\frac{b-\beta_1}{b}$)	75.94%		68.08%	75.64%	78.01%	77.99%

A. Model Appendix

A.1. Derivation of Bargaining Rules

Derivation of sharing rule for internal promotions. Suppose an executive is approached by their firm and evaluated for a promotion to CEO. First, a new match productivity parameter p' is drawn, summarizing the executive's fitness for the CEO position. Each party makes alternating offers over the piece rate r' . If the offer is accepted, the bargaining game ends. If the offer is rejected, some time elapses before a counteroffer is made. Let Δ_e and Δ_f respectively denote the lengths of time which elapse following a rejection by the executive and firm. It is also assumed that during negotiations, the match severs at rate ξ in which case the executive remains in their current position, and additional offers for outside CEO and non-CEO positions respectively arrive at rates λ_1 and λ_2 . The subgame perfect equilibrium of this game consists of piece rate offers (r_e, r_f) which make the other party indifferent between immediate acceptance and waiting to make a counteroffer. That is, r_e and r_f respectively solve:

$$V_C(r_f, h, \tau, p') = \frac{1}{1 + \rho\Delta_e} \left[w_t\Delta_e + \xi\Delta_e V_E(r, h, \tau, p) + \lambda_1\Delta_e \tilde{V}_C(\cdot) + \lambda_2\Delta_e \tilde{V}_E(\cdot) + (1 - \Delta_e(\xi + \lambda_1 + \lambda_2))V_C(r_e, h, \tau, p') \right] \quad (\text{A.1})$$

$$\Pi_C(r_e, h, \tau, p') = \frac{1}{1 + \rho\Delta_f} \left[\pi_0\Delta_e + \xi\Delta_e \Pi_0 + \lambda_0\Delta_f \tilde{\Pi}_C(\cdot) + \lambda_1\Delta_f \tilde{\Pi}_C(\cdot) + (1 - \Delta_f(\xi + \lambda_0 + \lambda_1))\Pi_C(r_f, h, \tau, p') \right] \quad (\text{A.2})$$

$\Pi_C(x)$ denotes the value to the firm of filling the CEO position given state x . π_0 and Π_0 denote the flow and net present values to the firm of having a vacant CEO position, both of which we assume to equal 0. $\tilde{V}_C$ and $\tilde{V}_E$ denote the executive's net present value of initiating a new bargaining game for a CEO or non-CEO position upon the arrival of a competing offer. Similar for $\tilde{\Pi}_C$. The two equations above can be rewritten as:

$$V_C(r_f, h, \tau, p') - V_C(r_e, h, \tau, p') = -\Delta_e \left[(\xi + \lambda_1 + \lambda_2)V_C(r_e, h, \tau, p') + \rho V_C(r_f, h, \tau, p') \right]$$

$$-w_t - \xi V_E(r, h, \tau, p) - \lambda_1 \tilde{V}_C(\cdot) - \lambda_2 \tilde{V}_E(\cdot) \Big] \quad (\text{A.3})$$

$$\begin{aligned} \Pi_C(r_e, h, \tau, p') - \Pi_C(r_f, h, \tau, p') = -\Delta_f \Big[& (\xi + \lambda_0 + \lambda_1) \Pi_C(r_f, h, \tau, p') + \rho \Pi_C(r_e, h, \tau, p') \\ & - \pi_0 - \xi \Pi_0 - \lambda_0 \tilde{\Pi}_C(\cdot) - \lambda_1 \tilde{\Pi}_E(\cdot) \Big] \end{aligned} \quad (\text{A.4})$$

The above conditions imply that $r_f \rightarrow r_e$ as $\Delta_f, \Delta_e \rightarrow 0$. Denote their common limit by r' and define:

$$\frac{\partial V_C}{\partial r}(r, h, \tau, p') = \lim_{\Delta_f, \Delta_e \rightarrow 0} \frac{V_C(r_f, h, \tau, p') - V_C(r_e, h, \tau, p')}{r_f - r_e} \quad (\text{A.5})$$

$$\frac{\partial \Pi_C}{\partial r}(r, h, \tau, p') = \lim_{\Delta_f, \Delta_e \rightarrow 0} \frac{\Pi_C(r_f, h, \tau, p') - \Pi_C(r_e, h, \tau, p')}{r_f - r_e} \quad (\text{A.6})$$

Using the definitions above and taking the ratios of (A.3) and (A.4) yields:

$$-\frac{\frac{\partial V_C}{\partial r}(r', h, \tau, p')}{\frac{\partial \Pi_C}{\partial r}(r', h, \tau, p')} = \frac{\Delta_e(\rho + \xi + \lambda_1 + \lambda_2)}{\Delta_f(\rho + \xi + \lambda_0 + \lambda_1)} \frac{V_C(r', h, \tau, p') - \frac{w_t + \xi V_E(r, h, \tau, p) + \lambda_1 \tilde{V}_C(\cdot) + \lambda_2 \tilde{V}_E(\cdot)}{\rho + \xi + \lambda_1 + \lambda_2}}{\Pi_C(r', h, \tau, p') - \frac{\pi_0 + \xi \Pi_0 + \lambda_0 \tilde{\Pi}_C(\cdot) + \lambda_1 \tilde{\Pi}_E(\cdot)}{\rho + \xi + \lambda_0 + \lambda_1}} \quad (\text{A.7})$$

Next, define $S(h, \tau, p') = \Pi_C(r', h, \tau, p') + V_C(r', h, \tau, p') - V_E(r, h, \tau, p)$ as the surplus associated with the position. Note that $\Pi_C(0, h, \tau, p') = 0$, which implies that $\Pi_C(r', h, \tau, p') = V_C(0, h, \tau, p') - V_C(r', h, \tau, p')$. Thus, $\frac{\partial \Pi_C}{\partial r}(r', h, \tau, p') = -\frac{\partial V_C}{\partial r}(r', h, \tau, p')$. Applying this to (A.7) and taking the limit as $\xi \rightarrow \infty$ yields (after some algebra):

$$V_C(r', h, \tau, p') = \beta_0 V_C(0, h, \tau, p') + (1 - \beta_0) V_E(r, h, \tau, p) \quad (\text{A.8})$$

where $\beta_0 = \frac{\Delta_f}{\Delta_f + \Delta_e}$.¹ Note that the sharing rule implies that it is in the executive's interest to accept the promotion to CEO if and only if $V_C(0, h', \tau', p') > V_E(r, h', \tau', p)$. Define $\bar{\theta}_0(r, h, \tau, p)$ as

¹Under this game-theoretic interpretation of the bargaining parameter, the executive's bargaining power increases in the time Δ_f it takes the firm to come up with a counteroffer. This is fairly unimportant and in no way changes the empirical treatment of the parameter β_0 . Also, the form of the sharing rules hinges on the assumption that the separation rate ξ is large relative to the arrival rates of competing offers. This is a technical assumption that for our purposes will be buried in the appendix.

the critical level of match productivity such that:

$$V_C(0, h, \tau, \bar{\theta}_0) = V_E(r, h, \tau, p) \quad (\text{A.9})$$

If $p' > \bar{\theta}_0$, the executive accepts the CEO role under the wage contract defining (A.8). If $p' \leq \bar{\theta}_0$, the executive can never be swayed into accepting the CEO role and instead remains in their current position.

Derivation of sharing rule for horizontal moves (Executive). Consider an executive who is approached by an outside firm to serve in an executive position. Upon the arrival of the offer, the executive along with the competing and incumbent firms initiate a bargaining game with the following structure:

1. Stage 1: Both firms simultaneously offer a piece rate to the executive
2. Stage 2: The executive chooses one of the offers, or rejects and keeps their current position.
3. Stage 3: If the executive accepted an offer in Stage 2, some time elapses. The executive then renegotiates with the firm whose offer was rejected, where the renegotiation protocol mirrors that of the previous section. Unlike the previous section, however, the executive's outside option is not their current position, but the offer accepted in Stage 2.

The bargaining game is solved via backward induction. Let r'_1 and r_1 denote the Stage 1 offers from firms p' and p . Suppose that r_1 was accepted in the second stage, triggering a Stage 3 renegotiation with firm p' . The offer from p' will satisfy:

$$V_E(r, h, 0, p') = \beta_1 V_E(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p) \quad (\text{A.10})$$

Conversely, suppose that r'_1 was accepted in Stage 2. In the subsequent renegotiation with firm p , their counteroffer will satisfy:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(r'_1, h, 0, p') \quad (\text{A.11})$$

The form of the counteroffers implies that:

- If r'_1 was accepted in Stage 2, renegotiate and eventually work with p iff:

$$V_E(0, h, \tau, p) \geq V_E(r'_1, h, 0, p') \quad (\text{A.12})$$

- If r_1 was accepted in Stage 2, renegotiate and eventually work with p' iff:

$$V_E(0, h, 0, p') > V_E(r_1, h, \tau, p) \quad (\text{A.13})$$

Thus, the value of accepting r_1 at Stage 2 is:

$$V = \max\{\beta_1 V_E(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p), V_E(r_1, h, \tau, p)\} \quad (\text{A.14})$$

Similarly for r'_1 :

$$V = \max\{\beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(r'_1, h, 0, p'), V_E(r'_1, h, 0, p')\} \quad (\text{A.15})$$

Moving back to Stage 1, both firms make simultaneous offers. For firm p' to eventually win the executive, they must bid r'_1 such that $V_E(r'_1, h, 0, p') > V_E(0, h, \tau, p)$ so that firm p cannot afford to outbid p' . Acknowledging the transferability of p and k , firm p' eventually wins the worker if and only if $p' > p + k'$. In this case, to avoid wasting time in the renegotiation stage, firm p' immediately offers r'_1 such that:

$$V_E(r'_1, h, 0, p') = \beta_1 V_E(0, h, 0, p') + (1 - \beta_1) V_E(0, h, \tau, p) \quad (\text{A.16})$$

Conversely, if $p' < p + k'$, firm p will eventually win the executive's services. The fastest way of doing so is to immediately offer r_1 such that:

$$V_E(r_1, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(0, h, 0, p') \quad (\text{A.17})$$

Note additionally that a competing offer does not necessitate an alteration of the current piece rate r . The minimal value of p' such that something happens is defined by:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(0, h, 0, \underline{\theta}_E) \quad (\text{A.18})$$

Derivation of sharing rule for external promotions. Consider an executive who is approached by an outside firm to become CEO. A three-player bargaining game is initiated with the same structure as in the previous case. Note, however, that unlike the previous case, the executive is weighing two separate types of positions: a CEO position and a non-CEO position. Because the two position types are associated with different event spaces describing the possible set of future offers, the executive's value of accepting these positions, for a given state, is not equal in general.

As before, the bargaining game is solved via backward induction. Let $(r'_1, 1)$ and $(r_1, 0)$ denote the stage 1 offers from firms p' and p , where the second coordinate indicates if the offer is for a CEO position or not. Suppose the executive accepted $(r_1, 0)$ at Stage 2, then renegotiates with p' in Stage 3. The offer from p' will satisfy:

$$V_C(r, h, 0, p') = \beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p) \quad (\text{A.19})$$

Conversely, suppose that $(r'_1, 1)$ was accepted at stage 2, and a stage 3 renegotiation was triggered with firm p . Firm p 's counteroffer will satisfy:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(r'_1, h, 0, p') \quad (\text{A.20})$$

Implications:

- If $(r'_1, 1)$ is accepted at stage 2, renegotiate and eventually work with p iff:

$$V_E(0, h, \tau, p) \geq V_C(r'_1, h, 0, p') \quad (\text{A.21})$$

- If $(r_1, 0)$ is accepted at stage 2, renegotiate and eventually work with p' iff:

$$V_C(0, h, 0, p') > V_E(r_1, h, \tau, p) \quad (\text{A.22})$$

Thus, the value of accepting $(r_1, 0)$ at stage 2 is:

$$V = \max\{\beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p), V_E(r_1, h, \tau, p)\} \quad (\text{A.23})$$

Similarly for $(r'_1, 1)$:

$$V = \max\{\beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(r'_1, h, 0, p'), V_C(r'_1, h, 0, p')\} \quad (\text{A.24})$$

At stage 1, simultaneous offers are made. For firm p' to win the executive, they must bid r'_1 such that: $V_C(r'_1, h, 0, p') > V_E(0, h, \tau, p) = V_C(0, h, \tau, \bar{\theta}_0)$. Again acknowledging the transferability of k and p , the previous inequality is equivalent to $V_C(r'_1, h', 0, p') > V_C(0, h', 0, \bar{\theta}_0 + k')$. Hence, firm p' eventually wins the worker if and only if $p' > \bar{\theta}_0 + k'$. In this case, to avoid wasting time in the renegotiation stage, firm p' immediately offers r'_1 such that:

$$V_C(r'_1, h, 0, p') = \beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(0, h, \tau, p) \quad (\text{A.25})$$

Conversely, suppose that $p' < \bar{\theta}_0 + k'$. Similar to the above case, firm p will retain the worker in the fastest manner possible by immediately offering r_1 such that:

$$V_E(r_1, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, p') \quad (\text{A.26})$$

As in the case of horizontal moves, an outside offer for a CEO appointment need not trigger a change in the current piece rate r . The minimum value of p' such that something happens is defined by:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, \underline{\theta}_O) \quad (\text{A.27})$$

Derivation of sharing rule for horizontal moves (CEO). This case is identical to the case for horizontal executive moves if we simply change subscripts. Upon receiving an offer for a CEO position with match productivity $p' > p+k$, the CEO switches positions and receives initial piece rate r' defined by condition:

$$V_E(r', h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, p') \quad (\text{A.28})$$

As in the previous cases, if instead $p' > p+k$, the current firm retains the CEO and a renegotiation may be triggered. The minimum value of p' such that the piece rate is revised is defined by:

$$V_C(r, h, \tau, p) = \beta_1 V_C(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, \underline{\theta}_C) \quad (\text{A.29})$$

A.2. Value Function Derivation

To obtain closed-form expressions for managerial piece rates, we first derive the executive and CEO value function, and then use the value functions to derive the piece rates. We assume that managers have logarithmic flow utility and that there is no transfer of wealth across time. All parties discount the future at rate ρ . Let $S(\cdot) = 1 - F(\cdot)$ be the survivor function for the distribution of position types. Given the threshold position $\underline{\theta}$ which leads to no contract revision, $S(\underline{\theta})$ represents the fraction of positions for which the manager discards the job offer.

A.2.1. CEO Value Function

The CEO value function is

$$\begin{aligned}
(\rho + \mu + \eta + \lambda_1 S(\underline{\theta}_C))V_C(r, h, k, p) = & w + \eta V_U(h) + \\
& \lambda_1 \int_{p+k}^{p_{\max}} \mathbb{E} \left\{ (1 - \beta_1)V_C(0, h, k, p) + \beta_1 V_C(0, h', 0, x) \right\} dF(x) + \\
& \lambda_1 \int_{\underline{\theta}_C}^{p+k} \mathbb{E} \left\{ (1 - \beta_1)V_C(0, h', 0, x) + \beta_1 V_C(0, h, k, p) \right\} dF(x)
\end{aligned}$$

The net present value of holding a CEO position is the sum of flow compensation and expectations over future employment transitions. Note that the integral terms in the value function above reflect the structure of the CEO bargaining process.

Rearranging terms via integration by parts yields:

$$\begin{aligned}
(\rho + \mu + \eta)V_C(r, k, p) = & r + g + k + p + \eta V_U(h) + \lambda_1 \beta_1 \int_{p+k}^{p_{\max}} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx \\
& + \lambda_1 (1 - \beta_1) \int_{\underline{\theta}_C}^{p+k} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx
\end{aligned} \tag{A.30}$$

Setting $r = 0$ and differentiating with respect to p (applying Leibniz' rule) yields:

$$\frac{\partial V_C}{\partial p}(0, k, p) = \frac{1}{\rho + \mu + \eta + \lambda_1 \beta_1 S(p)} \equiv \phi_C(p) \tag{A.31}$$

The value function (A.30) can then be expressed as:

$$\begin{aligned}
(\rho + \mu + \eta)V_C(r, k, p) = & r + g + k + p + \eta V_U(h) + \lambda_1 \beta_1 \int_{p+k}^{p_{\max}} \phi_C(x) S(x) dx \\
& + \lambda_1 (1 - \beta_1) \int_{\underline{\theta}_C}^{p+k} \phi_C(x) S(x) dx
\end{aligned} \tag{A.32}$$

A.2.2. Executive Value Function

The net present value of holding an executive position, though slightly more cumbersome, is similar in structure:

$$\begin{aligned}
 \left(\rho + \mu + \eta + \sum_{s=0}^2 \lambda_s S(\underline{\theta}_s) \right) V_E(r, h, k, p) = & w + \eta V_U(h) + \\
 & \lambda_0 \int_{\bar{\theta}}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_0) V_E(0, h, k, p) + \beta_0 V_C(0, h, k, x) \right\} dF(x) + \\
 & \lambda_1 \int_{\bar{\theta}+k}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_1) V_E(0, h', k', p) + \beta_1 V_C(0, h', 0, x) \right\} dF(x) + \\
 & \lambda_1 \int_{\underline{\theta}_O}^{\bar{\theta}+k} \mathbb{E} \left\{ (1 - \beta_1) V_C(0, h', 0, x) + \beta_1 V_E(0, h', k', p) \right\} dF(x) + \\
 & \lambda_2 \int_{p+k}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_1) V_E(0, h', k', p) + \beta_1 V_E(0, h', 0, x) \right\} dF(x) + \\
 & \lambda_2 \int_{\underline{\theta}_2}^{p+k} \mathbb{E} \left\{ (1 - \beta_1) V_E(0, h', 0, x) + \beta_1 V_E(0, h', k', p) \right\} dF(x)
 \end{aligned} \tag{A.33}$$

Similar to the case of CEOs, we simplify the executive value function by first rearranging (A.33) via integration by parts:

$$\begin{aligned}
 (\rho + \mu + \eta) V_E(r, k, p) = & r + g + k + p + \eta V_U(h) + \lambda_0 \beta_0 \int_{\bar{\theta}}^{p_{max}} \frac{\partial V_C}{\partial x}(0, k, x) S(x) dx \\
 & + \lambda_1 \beta_1 \int_{\bar{\theta}+k}^{p_{max}} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx + \lambda_1 (1 - \beta_1) \int_{\underline{\theta}_O}^{\bar{\theta}+k} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx \\
 & + \lambda_2 \beta_1 \int_{p+k}^{p_{max}} \frac{\partial V_E}{\partial x}(0, 0, x) S(x) dx + \lambda_2 (1 - \beta_1) \int_{\underline{\theta}_E}^{p+k} \frac{\partial V_E}{\partial x}(0, 0, x) S(x) dx
 \end{aligned} \tag{A.34}$$

Again setting $r = 0$ and differentiating with respect to p yields:

$$\frac{\partial V_E}{\partial p}(0, k, p) = \frac{1 - \phi_C(\bar{\theta} + k) \frac{\partial \bar{\theta}}{\partial p} \left(\lambda_0 \beta_0 S(\bar{\theta}) + \lambda_1 \beta_1 S(\bar{\theta} + k) \right)}{\rho + \mu + \eta + \lambda_2 \beta_1 S(p + k)} \tag{A.35}$$

Differentiating Equation (5) evaluated at $r = 0$ yields:

$$\frac{\partial \bar{\theta}}{\partial p}(0, k, p) = \frac{\frac{\partial V_E}{\partial p}(0, k, p)}{\phi_C(\bar{\theta} + k)} \quad (\text{A.36})$$

Substituting the expression above into Equation (A.35) and simplifying then yields:

$$\frac{\partial V_E}{\partial p}(0, k, p) = \frac{1}{\rho + \mu + \eta + \lambda_2 \beta_1 S(p + k) + \lambda_1 \beta_1 S(\bar{\theta} + k) + \lambda_0 \beta_0 S(\bar{\theta})} \equiv \phi_E(p, k) \quad (\text{A.37})$$

The executive value function can then be expressed as:

$$\begin{aligned} (\rho + \mu + \eta)V_E(r, k, p) &= r + g + k + p + \eta V_U(h) + \lambda_0 \beta_0 \int_{\bar{\theta}}^{p_{\max}} \phi_C(x + k)S(x)dx \\ &\quad + \lambda_1 \beta_1 \int_{\bar{\theta} + k}^{p_{\max}} \phi_C(x)S(x)dx + \lambda_1(1 - \beta_1) \int_{\underline{\theta}_O}^{\bar{\theta} + k} \phi_C(x)S(x)dx \\ &\quad + \lambda_2 \beta_1 \int_{p + k}^{p_{\max}} \phi_E(x, 0)S(x)dx + \lambda_2(1 - \beta_1) \int_{\underline{\theta}_E}^{p + k} \phi_E(x, 0)S(x)dx \end{aligned} \quad (\text{A.38})$$

With both the executive and CEO value functions in hand, we can present the formal derivation of $\bar{\theta}(r, t, \tau, p)$.

Proof of Theorem 1. Observe that for all values of p :

$$\frac{\partial \bar{\theta}}{\partial p}(0, k, p) = \frac{\phi_C(p, k)}{\phi_E(\bar{\theta} + k)} = \frac{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(\bar{\theta} + k)}{\rho + \mu + \eta + \lambda_2 \beta_1 \bar{F}(p + k) + \lambda_1 \beta_1 \bar{F}(\bar{\theta} + k) + \lambda_0 \beta_0 \bar{F}(\bar{\theta})} \leq 1 \quad (\text{A.39})$$

Then:

$$\begin{aligned} p - \bar{\theta}(0, k, p) &= \int_{p_{\min}}^p dx - \int_{p_{\min}}^p \frac{\partial \bar{\theta}}{\partial x}(0, k, x)dx \\ &= \int_{p_{\min}}^p \left(1 - \frac{\partial \bar{\theta}}{\partial x}(0, k, x) \right) dx \geq 0 \end{aligned} \quad (\text{A.40})$$

since the integrand is positive for all values in the support of p . ■

A.3. Piece Rate Derivation

A.3.1. CEO piece rates

Horizontal Hires. The bargaining condition (A.28) implies:

$$\begin{aligned} r = & \beta_1 \left(p' + \lambda_1 \beta_1 \int_{p'}^{p_{\max}} \phi_C(x) S(x) dx \right) + (1 - \beta_1) \left(p + k_- + \lambda_1 \beta_1 \int_{p+k_-}^{p_{\max}} \phi_C(x) S(x) dx \right) \\ & - p' - \lambda_1 \beta_1 \int_{p'}^{p_{\max}} \phi_C(x) S(x) dx - \lambda_1 (1 - \beta_1) \int_{p+k_-}^{p'} \phi_C(x) S(x) dx \end{aligned} \quad (\text{A.41})$$

Combining terms yields:

$$\begin{aligned} r = & -(1 - \beta_1)(p' - p - k_-) - \lambda_1 (1 - \beta_1)^2 \int_{p+k_-}^{p'} \phi_C(x) S(x) dx \\ = & -(1 - \beta_1) \int_{p+k_-}^{p'} \frac{\rho + \mu + \eta + \lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 S(x)} dx \\ = & \int_{p+k_-}^{p'} q(x) dx \equiv r_C^{\text{hor}}(p', 0, p + k_-) \end{aligned} \quad (\text{A.42})$$

Diagonal Hires. Applying the definition of $\bar{\theta}$ (Equation (5)), the bargaining condition (A.26) can be rewritten as:

$$V_C(r, 0, p) = \beta_1 V_C(0, 0, p') + (1 - \beta_1) V_C(0, k, \bar{\theta}(0, k, p)) \quad (\text{A.43})$$

Inserting the associated value functions then yields:

$$\begin{aligned} r = & \beta_1 \left(p' + \lambda_1 \beta_1 \int_{p'}^{p_{\max}} \phi_C(x) S(x) dx \right) + (1 - \beta_1) \left(\bar{\theta}(p, k_-) + k_- + \lambda_1 \beta_1 \int_{p+k_-}^{p_{\max}} \phi_C(x) S(x) dx \right) \\ & - p' - \lambda_1 \beta_1 \int_{p'}^{p_{\max}} \phi_C(x) S(x) dx - \lambda_1 (1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} \phi_C(x) S(x) dx \end{aligned} \quad (\text{A.44})$$

Collecting terms and rearranging yields:

$$r = -(1 - \beta_1)(p' - \bar{\theta}(p, k_-) - k_-) - \lambda_1 (1 - \beta_1)^2 \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} \phi_C(x) S(x) dx \quad (\text{A.45})$$

$$= -(1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} \frac{\rho + \mu + \eta + \lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 S(x)} dx \quad (\text{A.46})$$

$$= -(1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} q(x) dx \quad (\text{A.47})$$

$$= r_C^{hor}(p', 0, p + k_-) - (1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p + k_-} q(x) dx \equiv r_C^{diag}(p', 0, p + k_-) \quad (\text{A.48})$$

Vertical Hires. Similar to the previous case, we can apply the definition of $\bar{\theta}$ and rewrite the bargaining condition (A.8) as:

$$V_C(r, k, p') = \beta_0 V_C(0, k, p') + (1 - \beta_0) V_C(0, k, \bar{\theta}(r, k, p)) \quad (\text{A.49})$$

Inserting the definition of the value function yields:

$$\begin{aligned} r = & \beta_0 \left(p' + k + \lambda_1 \beta_1 \int_{p'}^{p_{max}} \phi_C(x) S(x) dx \right) + (1 - \beta_0) \left(\bar{\theta}(r, k, p) + k + \lambda_1 \beta_1 \int_{p + k_-}^{p_{max}} \phi_C(x) S(x) dx \right) \\ & - p' - k - \lambda_1 \beta_1 \int_{p'}^{p_{max}} \phi_C(x) S(x) dx - \lambda_1 (1 - \beta_1) \int_{\bar{\theta}(r, k, p) + k}^{p'} \phi_C(x) S(x) dx \end{aligned} \quad (\text{A.50})$$

Substituting the identity $\beta_0 = \beta_0 + \beta_1 - \beta_1$ and collecting terms yields:

$$\begin{aligned} r = & \overbrace{-(1 - \beta_1) \int_{\bar{\theta}(r, k, p) + k}^{p'} q(x) dx}^{r_C^{diag}(p', 0, p + k)} - (1 - \beta_1) \int_{p'}^{p' + k} q(x) dx \\ & + (\beta_0 - \beta_1) \int_{\bar{\theta}(r, k, p) + k}^{p'} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \equiv r_C^{vert}(p', 0, p + k) \end{aligned} \quad (\text{A.51})$$

A.3.2. Executive Piece Rate Derivation

We show that the executive piece rate, upon switching to a p' position from a p position with k_- firm-specific human capital can be written as:

$$r_E(p', 0, p + k_-) = -(1 - \beta_1) \left\{ \int_{p + k_-}^{p'} q_E(x) dx \right.$$

$$\left. -\lambda_0\beta_0 \int_{\bar{\theta}(p)}^{\bar{\theta}(p')} \phi_C(x)S(x) dx + \lambda_1(1-\beta_1) \int_{\bar{\theta}(p)+k_-}^{\bar{\theta}(p')} \phi_C(x)S(x) dx \right\} \quad (\text{A.52})$$

where

$$q_E(x) = \frac{\rho + \mu + \eta + \lambda_0\beta_0 S(\bar{\theta}(x)) + \lambda_1\beta_1 S(x) + \lambda_2 S(x)}{\rho + \mu + \eta + \lambda_0\beta_0 S(\bar{\theta}(x)) + \lambda_1\beta_1 S(x) + \lambda_2\beta_1 S(x)}$$

$$\phi_C(x) = \frac{1}{\rho + \mu + \eta + \lambda_1\beta_1 S(x)} = \frac{\partial V_C}{\partial x}(0, h, 0, x)$$

The bargaining condition for horizontal executive moves (Equation (A.16)) implies:

$$\begin{aligned} r = & \beta_1 \left(p' + \lambda_0\beta_0 \int_{\bar{\theta}(p',0)}^{p_{\max}} \phi_C(x)S(x)dx + \lambda_1\beta_1 \int_{\bar{\theta}(p,k_-)}^{p_{\max}} \phi_C(x)S(x)dx + \lambda_2\beta_1 \int_{p'}^{p_{\max}} \phi_E(x)S(x)dx \right) \\ & + (1-\beta_1) \left(p + k_- + \lambda_0\beta_0 \int_{\bar{\theta}(p,k_-)}^{p_{\max}} \phi_C(x)S(x)dx + \lambda_1\beta_1 \int_{\bar{\theta}(p,k_-)+k_-}^{p_{\max}} \phi_C(x)S(x)dx + \lambda_2\beta_1 \int_{p+k_-}^{p_{\max}} \phi_E(x)S(x)dx \right) \\ & - p' - \lambda_0\beta_0 \int_{\bar{\theta}(p',0)}^{p_{\max}} \phi_C(x)S(x)dx - \lambda_1\beta_1 \int_{\bar{\theta}(p,k_-)}^{p_{\max}} \phi_C(x)S(x)dx - \lambda_2\beta_1 \int_{p'}^{p_{\max}} \phi_E(x)S(x)dx \\ & - \lambda_1(1-\beta_1) \int_{\bar{\theta}(p,k_-)+k_-}^{\bar{\theta}(p',0)} \phi_C(x)S(x)dx - \lambda_2(1-\beta_1) \int_{p+k_-}^{p'} \phi_E(x)S(x)dx \end{aligned} \quad (\text{A.53})$$

Simplifying and collecting terms yields:

$$\begin{aligned} r = & -(1-\beta_1)(p' - p - k_-) - \lambda_2(1-\beta_1)^2 \int_{p+k_-}^{p'} \phi_E(x)S(x)dx \\ & - \lambda_1(1-\beta_1)^2 \int_{\bar{\theta}(p,k_-)+k_-}^{\bar{\theta}(p',0)} \phi_C(x)S(x)dx + \lambda_0\beta_0(1-\beta_1) \int_{\bar{\theta}(p,k_-)}^{\bar{\theta}(p',0)} \phi_C(x)S(x)dx \end{aligned} \quad (\text{A.54})$$

which is equivalent to Equation (A.52).

A.4. Characterization of the CEO Renegotiation Set

We restrict the analysis of this section to the special case of the model in which $\beta_0 = \beta_1$; more general results are a work in progress. The renegotiation set is defined as the closed interval $[\theta_C(p, k), p + k]$ with mass $\Omega_r^m(p, k)$ for $m \in \{hor, diag, vert\}$. $\theta_C(p, k)$ is defined implicitly by the

following equation:

$$r = - \int_{\theta_C(p,k)}^{p+k} q(x)dx \quad (\text{A.55})$$

where the left hand side denotes the CEO's current piece rate and the right hand side is the revised piece rate in case of a renegotiation. The form of the piece rate r , of course, depends on the path up the job ladder the manager took before landing in the CEO position. We begin with the case of horizontally-hired CEOs. Letting $p_- + k_-$ denote the sum of the match productivity and firm-specific human capital associated with the CEO's previous position at time of departure, inserting the definition of r_C^{hor} (Equation (13)), and rearranging terms yields:

$$\int_z^p q(x)dx = \int_{\theta_C(p,k)}^{p+k} q(x)dx \quad (\text{A.56})$$

$$\int_p^{p+k} q(x)dx = \int_{p_-+k_-}^{\theta_C(p,k)} q(x)dx \quad (\text{A.57})$$

Equation (A.57) implies that at time of hire (i.e. when $k = 0$), $\theta_C(p, 0) = p_- + k_-$. The dynamics of $\theta_C(p, k)$ as k increases can be obtained by implicitly differentiating (A.57) with respect to k :

$$\frac{\partial \theta_C}{\partial k}(p, k) = \frac{q(p+k)}{q(\theta_C(p, k))} < 1 \quad (\text{A.58})$$

where the inequality follows from the fact that $q(x)$ is monotonically decreasing. Additionally, the differential equation (A.58) is independent of $p_- + k_-$. Thus, while the initial value of θ_C upon taking a new CEO position will differ by CEO type, the slope of θ_C with respect to k is independent of CEO type. Using the same approach as above, the initial condition of θ_C for diagonally-hired CEOs is apparent by inspecting the equation:

$$\int_p^{p+k} q(x)dx = \int_{\bar{\theta}(p_-, k_-)+k_-}^{\theta_C(p,k)} q(x)dx \quad (\text{A.59})$$

Hence, for diagonally-hired CEOs $\theta_C(p, 0) = \bar{\theta}(p_-, k_-) + k_-$. Characterizing θ_C for vertically-hired CEOs is made difficult by the fact that $\beta_0 \neq \beta_1$ in general. We thus restrict attention to the special

case where $\beta_0 = \beta_1$ for now. Consider then a vertically-hired CEO with τ units of firm-specific tenure at time of hire. We can write:

$$\int_{\bar{\theta}(p_-,k)+k}^{p+k(\tau)} q(x)dx = \int_{\theta_C(p,k)}^{p+k(\tau+\Delta)} q(x)dx \quad (\text{A.60})$$

$$\int_{p+k(\tau)}^{p+k(\tau+\Delta)} q(x)dx = \int_{\bar{\theta}(p_-,k)+k}^{\theta_C(p,k)} q(x)dx \quad (\text{A.61})$$

At the point of promotion (when $\Delta = 0$), $\theta_C(p, k(\tau)) = \bar{\theta}(p_-, k) + k$. Because θ_C monotonically increases with k , this implies that the value of $\theta_C(p, 0)$ is smaller for vertically-hired CEOs relative to diagonally-hired CEOs. With these results in hand, we can state formally the proof of Theorem 2:

Proof of Theorem 2. Define $\Omega_r^m(p, k) = p + k - \theta_C(p, k)$. Differentiating with respect to p :

$$\frac{\partial \Omega_r^m}{\partial k}(p, k) = 1 - \frac{\partial \theta_C}{\partial k}(p, k) > 0 \quad (\text{A.62})$$

which follows from the fact that $\frac{\partial \theta_C}{\partial k}(p, k) < 1$. Let $z^m = \theta_C(p, 0)$ for $m \in \{hor, diag, vert\}$ denote the type-specific initial condition of $\theta_C(p, k)$. As $z^{vert} < z^{diag} < z^{hor}$, it follows that for any (p, k) :

$$\Omega_r^{vert}(p, k) > \Omega_r^{diag}(p, k) > \Omega_r^{hor}(p, k) \quad (\text{A.63})$$

■

We have

$$\begin{aligned} r_C^{vert}(p', 0, p' + k_-) &= r_C^{diag}(p', 0, p' + k_-) - (1 - \beta_1) \int_{p'}^{p'+k_-} q(x)dx \\ &\quad + (\beta_0 - \beta_1) \int_{\bar{\theta}(p)+k_-}^{p'} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \end{aligned} \quad (\text{A.64})$$

The proof holds trivially if $\beta_0 = \beta_1$ or $\beta_1 = 1$. Suppose $\beta_0 \leq \beta_1$. Consider an offer to be internally promoted to CEO such that $p' = \bar{\theta}$, i.e. the threshold offer. Any other accepted offer will be better.

Then, we need to show that

$$-(1 - \beta_1) \int_{\bar{\theta}}^{\bar{\theta}+k_-} q(x) dx + (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta}+k_-} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \leq 0$$

We can write

$$-(1 - \beta_1) \int_{\bar{\theta}}^{\bar{\theta}+k_-} q(x) dx + (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta}+k_-} q(x) dx - (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta}+k_-} \frac{\lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx$$

The third term is ≤ 0 , so we only need show

$$-(1 - \beta_1) \int_{\bar{\theta}}^{\bar{\theta}+k_-} q(x) dx + (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta}+k_-} q(x) dx \leq 0$$

This holds surely if $\beta_1 \leq \frac{\beta_0+1}{2}$. So, we have the statement:

If $\beta_0 \leq \beta_1 \leq \frac{\beta_0+1}{2}$ then vertically hired are always weakly smaller contracts. This holds for all values $\{(\beta_0, \beta_1) \in [0, 0.5]^2 \text{ s.t. } \beta_0 \leq \beta_1\}$.

However, the sharp bound is

$$(\beta_1 - \beta_0)[1 - A(r, p, k_-)] \leq (1 - \beta_1)$$

where

$$A(r, p, k_-) = \frac{\int_{\bar{\theta}(r, p, k_-)}^{\bar{\theta}(r, p, k_-)+k_-} v(x) dx}{\int_{\bar{\theta}(r, p, k_-)}^{\bar{\theta}(r, p, k_-)+k_-} q(x) dx} \leq 1$$

and $v(x) = \lambda_1 S(X)/(\rho + \mu + \eta + \lambda_1 \beta_1 S(x))$

B. Estimation Appendix

B.1. Weighting Matrix

From the empirical sample we obtain a $K \times 1$ vector of moments $\hat{M}$. Let Ψ denote the corresponding $N \times K$ matrix of influence functions, N being the number of observations in the sample. Each element Ψ_{nk} is the influence function describing observation n 's contribution to moment k . The

covariance matrix of the vector of moments can then be estimated as:

$$avar(\hat{M}) = \Psi' \Psi \quad (\text{B.1})$$

The weighting matrix $\hat{W}$ is then obtained as the inverse of matrix [B.1](#). Let $\Theta \in \mathbb{R}^P$ denote an arbitrary vector of structural parameters. Define the moment residual $g : \mathbb{R}^P \rightarrow \mathbb{R}^M$ as:

$$g(\Theta) = \hat{M} - \frac{1}{S} \sum_{s=1}^S \hat{m}^s(\Theta) \quad (\text{B.2})$$

Where $\hat{M}$ is the vector of empirical moments, $\hat{m}^s(\Theta)$ is the vector of simulated moments given parameter values Θ in simulation s , and S is the total number of simulations. The vector of estimates $\hat{\Theta}$ minimizes the SMM objective function:

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta) \hat{W} g(\Theta)' \quad (\text{B.3})$$

B.2. Model Estimation Algorithm

We minimize the SMM quadratic form using the TikTak algorithm. The routine proceeds as follows:

1. *Set initial guesses for model parameters:*
2. *Simulate model:*
3. *Construct simulated panel and compute moments:* Using the simulated data, we construct a panel resembling the empirical sample and compute the same moments as described in [Section 4](#).
4. *Evaluate objective function:*

B.3. Standard Errors for Parameter Estimates

For true parameter vector Θ and consistent estimate $\hat{\Theta}$, we have the following asymptotic distribution (Duffie and Singleton, 1993):

$$\sqrt{n}(\hat{\Theta} - \Theta) \rightarrow^d N(0, avar(\hat{\Theta})) \quad (\text{B.4})$$

$avar(\hat{\Theta})$ can be expressed as:

$$avar(\hat{\Theta}) = \left(1 + \frac{1}{S}\right) \left(\frac{\partial g(\Theta)}{\partial \Theta} W \frac{\partial g(\Theta)}{\partial \Theta'} \right)^{-1} \quad (\text{B.5})$$

where $\frac{\partial g(\Theta)}{\partial \Theta}$ is the Jacobian of the moment residual (B.2) with respect to the structural parameters, W is the optimal weighting matrix, and S is the number of simulations. We approximate the Jacobian using:

$$\frac{\partial \hat{g}_m(\Theta)}{\partial \Theta_p} = \frac{g_p(\hat{\Theta} + h_p) - g_p(\hat{\Theta})}{h_p} \quad (\text{B.6})$$

for each moment m and parameter p . h_p is the perturbation size for parameter which we set to 1% of the absolute value of the parameter estimate. The standard errors are then obtained as the square root of the diagonal elements of the matrix:

$$\left(1 + \frac{1}{S}\right) \left(\frac{\partial \hat{g}(\Theta)}{\partial \Theta} \hat{W} \frac{\partial \hat{g}(\Theta)}{\partial \Theta'} \right)^{-1} \quad (\text{B.7})$$

where $\hat{W}$ is the sample counterpart of the optimal weighting matrix.